



```

name: The value of a helping hand
log: C:\Users\hansg\Documents\working\SHARE data\weighting\output_v6_1.smcl
log type: smcl
opened on: 9 Jun 2026, 19:01:31
C:\Users\hansg\Documents\working\SHARE data\weighting
(1 observation deleted)
(10,379 real changes made)
(6,150 real changes made)
(70,210 real changes made)
(8,277 real changes made)
(23,812 real changes made)
(1,763 real changes made)
(1,200 real changes made)
(1,763 real changes made)

```

Skewness and kurtosis tests for normality

Variable	Obs	Pr(skewness)	Pr(kurtosis)	Joint test	
				Adj chi2(2)	Prob>chi2
age	109,546	0.0000	0.0000	.	.

Skewness and kurtosis tests for normality

Variable	Obs	Pr(skewness)	Pr(kurtosis)	Joint test	
				Adj chi2(2)	Prob>chi2
thinc	109,546	0.0000	0.0000	.	.

Skewness and kurtosis tests for normality

Variable	Obs	Pr(skewness)	Pr(kurtosis)	Joint test	
				Adj chi2(2)	Prob>chi2
yedu	109,546	0.0000	0.0000	106.19	0.0000
(11,207 real changes made)					

Summary for variables: id

Group variable: htype (Household type)

htype	N
1 S	37717
1 CNRP	8553
1 C2R	62737
1 CNRP + 1 S	17
Multi S	372
1 C2R + 1 S	117
1 C2R + 1 CNRP	2
Multi C2R	29
Multi S + 1 C2R	2
Total	109546

Household type	Freq.	Percent	Cum.
1	37,717	34.43	34.43
2	8,553	7.81	42.24
3	62,737	57.27	99.51
4	17	0.02	99.52
5	372	0.34	99.86
6	117	0.11	99.97
7	2	0.00	99.97
8	29	0.03	100.00
10	2	0.00	100.00
Total	109,546	100.00	

(37,717 real changes made)
(62,737 real changes made)

Panel variable: **id** (unbalanced)
 Time variable: **Qyear, 2011 to 2022**, but with gaps
 Delta: **1 unit**
 (File stats_1.doc already exists, option **append** was assumed)

	Mean	SD	Min	Max	N
sphus	3.240347	1.029994	1	5	109546
age	73.86319	7.675333	59	100	109546
yedu	10.35106	4.508942	0	30	109546
thinc	27489.21	24343.79	0	93998.21	109546
gender	1.560687	.4963057	1	2	109546
household	.7386942	.5991849	0	2	109546
cjs	1.476667	1.271666	0	6	109546
covid	.1023041	.3030491	0	1	109546
Qyear	2014.987	3.64214	2011	2022	109546
id	14872.16	8611.134	1	29995	109546
gali	.52357	.4994464	0	1	109546
rhfo	.3778869	.6214552	0	2	109546
ghfo	.6235828	.8182134	0	2	109546
lifesat	7.811896	1.705112	0	10	109546
naly	.8736513	.377589	0	2	109546
country	18.88844	6.459957	11	34	109546

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		Year				
		2011	2013	2015	2017	2020
>						
>	2022 Total					
Self-perceived health - US scale						
>	Excellent	6.828	6.144	5.467	4.993	5.005
>	4.595 5.775					
>	Very good	16.433	15.777	15.575	14.168	15.173
>	14.161 15.459					
>	Good	37.576	38.797	37.986	39.700	40.738
>	40.671 38.842					
>	Fair	28.008	28.187	29.533	29.350	29.300
>	29.669 28.799					
>	Poor	11.155	11.096	11.440	11.789	9.785
>	10.904 11.125					
>	Total	100.000	100.000	100.000	100.000	100.000
>	100.000 100.000					

(collection **Table** exported to file [perc1_1.xls](#))

			Year			
			2011	2013	2015	2017
>						
>	2020	2022 Total				
Self-perceived health - US scale						
>	Excellent					
>	Received help from others (how many)=None		0.877	0.889	0.821	0.816
>	0.806 0.808 0.853					
>	Received help from others (how many)=At least 1		0.123	0.111	0.179	0.184
>	0.194 0.192 0.147					
>	Very good					
>	Received help from others (how many)=None		0.866	0.868	0.808	0.784
>	0.782 0.788 0.833					
>	Received help from others (how many)=At least 1		0.134	0.132	0.192	0.216

```

> 0.218 0.212 0.167
Good
>
> Received help from others (how many)=None | 0.837 0.843 0.791 0.755
> 0.739 0.716 0.799
> Received help from others (how many)=At least 1 | 0.163 0.157 0.209 0.245
> 0.261 0.284 0.201
Fair
>
> Received help from others (how many)=None | 0.753 0.747 0.669 0.675
> 0.643 0.645 0.706
> Received help from others (how many)=At least 1 | 0.247 0.253 0.331 0.325
> 0.357 0.355 0.294
Poor
>
> Received help from others (how many)=None | 0.610 0.601 0.523 0.530
> 0.505 0.501 0.563
> Received help from others (how many)=At least 1 | 0.390 0.399 0.477 0.470
> 0.495 0.499 0.437
Total
>
> Received help from others (how many)=None | 0.796 0.796 0.729 0.716
> 0.698 0.686 0.755
> Received help from others (how many)=At least 1 | 0.204 0.204 0.271 0.284
> 0.302 0.314 0.245

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(collection **Table** exported to file propalexcl_1.xls)

				Year				
				2011	2013	2015	2017	202
>	0	2022	Total					
Self-perceived health - US scale								
Excellent								
>			Given help to others (how many)=None	0.661	0.606	0.640	0.557	0.58
> 4	0.660	0.627	Given help to others (how many)=At least 1	0.339	0.394	0.360	0.443	0.41
> 6	0.340	0.373	Very good					
>			Given help to others (how many)=None	0.686	0.677	0.653	0.642	0.66
> 6	0.718	0.674	Given help to others (how many)=At least 1	0.314	0.323	0.347	0.358	0.33
> 4	0.282	0.326	Good					
>			Given help to others (how many)=None	0.744	0.732	0.726	0.716	0.73
> 8	0.778	0.739	Given help to others (how many)=At least 1	0.256	0.268	0.274	0.284	0.26
> 2	0.222	0.261	Fair					
>			Given help to others (how many)=None	0.785	0.790	0.809	0.803	0.81
> 8	0.850	0.806	Given help to others (how many)=At least 1	0.215	0.210	0.191	0.197	0.18
> 2	0.150	0.194	Poor					
>			Given help to others (how many)=None	0.874	0.884	0.898	0.902	0.91
> 7	0.901	0.892	Given help to others (how many)=At least 1	0.126	0.116	0.102	0.098	0.08
> 3	0.099	0.108	Total					
>			Given help to others (how many)=None	0.756	0.749	0.754	0.740	0.76

```

> 0    0.799    0.759
    Given help to others (how many)=At least 1 |    0.244    0.251    0.246    0.260    0.24
> 0    0.201    0.241

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(collection **Table** exported to file prop2excl_1.xls)

				2011	2013	2015	20
Self-perceived health - US scale							
Excellent							
Received help from others (how many)=None				0.877	0.889	0.821	0.4
60	0.806	0.808	0.806	0.123	0.111	0.179	0.1
Received help from others (how many)=At least 1				0.000	0.000	0.000	0.4
04	0.194	0.192	0.138	0.000	0.000	0.000	0.4
Received help from others (how many)=Not applicable							
36	0.000	0.000	0.056				
Very good							
Received help from others (how many)=None				0.866	0.868	0.808	0.4
40	0.782	0.788	0.783	0.134	0.132	0.192	0.1
Received help from others (how many)=At least 1				0.000	0.000	0.000	0.4
22	0.218	0.212	0.157	0.000	0.000	0.000	0.4
Received help from others (how many)=Not applicable							
38	0.000	0.000	0.059				
Good							
Received help from others (how many)=None				0.837	0.843	0.791	0.3
56	0.739	0.716	0.735	0.163	0.157	0.209	0.1
Received help from others (how many)=At least 1				0.000	0.000	0.000	0.5
16	0.261	0.284	0.185	0.000	0.000	0.000	0.5
Received help from others (how many)=Not applicable							
28	0.000	0.000	0.080				
Fair							
Received help from others (how many)=None				0.753	0.747	0.669	0.3
22	0.643	0.645	0.650	0.247	0.253	0.331	0.1
Received help from others (how many)=At least 1				0.000	0.000	0.000	0.5
55	0.357	0.355	0.271	0.000	0.000	0.000	0.5
Received help from others (how many)=Not applicable							
23	0.000	0.000	0.079				
Poor							
Received help from others (how many)=None				0.610	0.601	0.523	0.2
40	0.505	0.501	0.515	0.390	0.399	0.477	0.2
Received help from others (how many)=At least 1				0.000	0.000	0.000	0.5
13	0.495	0.499	0.399	0.000	0.000	0.000	0.5
Received help from others (how many)=Not applicable							
47	0.000	0.000	0.086				
Total							
Received help from others (how many)=None				0.796	0.796	0.729	0.3
50	0.698	0.686	0.698	0.204	0.204	0.271	0.1
Received help from others (how many)=At least 1				0.000	0.000	0.000	0.5
39	0.302	0.314	0.227	0.000	0.000	0.000	0.5
Received help from others (how many)=Not applicable							
11	0.000	0.000	0.076				

(collection **Table** exported to file prop1_1.xls)

				Year				
				2011	2013	2015	2017	
>	2020	2022	Total					
Self-perceived health - US scale								
> Excellent								
> Given help to others (how many)=None				0.460	0.434	0.640	0.314	
>	0.584	0.660	0.491	Given help to others (how many)=At least 1	0.236	0.282	0.360	0.250
>	0.416	0.340	0.292	Given help to others (how many)=Not applicable	0.304	0.284	0.000	0.436
>	0.000	0.000	0.217	Very good				
> Given help to others (how many)=None				0.469	0.482	0.653	0.361	
>	0.666	0.718	0.531	Given help to others (how many)=At least 1	0.215	0.231	0.347	0.201
>	0.334	0.282	0.257	Given help to others (how many)=Not applicable	0.316	0.287	0.000	0.438
>	0.000	0.000	0.212	Good				
> Given help to others (how many)=None				0.516	0.518	0.726	0.338	
>	0.738	0.778	0.575	Given help to others (how many)=At least 1	0.177	0.190	0.274	0.134
>	0.262	0.222	0.203	Given help to others (how many)=Not applicable	0.307	0.292	0.000	0.528
>	0.000	0.000	0.222	Fair				
> Given help to others (how many)=None				0.565	0.576	0.809	0.383	
>	0.818	0.850	0.637	Given help to others (how many)=At least 1	0.155	0.153	0.191	0.094
>	0.182	0.150	0.154	Given help to others (how many)=Not applicable	0.280	0.271	0.000	0.523
>	0.000	0.000	0.209	Poor				
> Given help to others (how many)=None				0.615	0.617	0.898	0.408	
>	0.917	0.901	0.687	Given help to others (how many)=At least 1	0.088	0.081	0.102	0.045
>	0.083	0.099	0.083	Given help to others (how many)=Not applicable	0.297	0.302	0.000	0.547
>	0.000	0.000	0.230	Total				
> Given help to others (how many)=None				0.529	0.535	0.754	0.361	
>	0.760	0.799	0.594	Given help to others (how many)=At least 1	0.171	0.179	0.246	0.127
>	0.240	0.201	0.189	Given help to others (how many)=Not applicable	0.300	0.286	0.000	0.511
>	0.000	0.000	0.217					

(collection **Table** exported to file prop2_1.xls)

		Current job situation				
>		Not applicable	Retired	Employed or self-employed	Unemployed	Permane
>	ntly sick or disabled	Homemaker	Other			
Year						
>						
>	2011	0.008	0.759		0.088	0.011
>		0.019	0.103	0.011		
>	2013	0.014	0.807		0.058	0.006
>		0.015	0.090	0.010		
>	2015	0.016	0.842		0.034	0.003
>		0.010	0.082	0.013		
>	2017	0.023	0.861		0.020	0.000
>		0.008	0.077	0.011		
>	2020	0.025	0.881		0.014	0.000
>		0.006	0.063	0.010		
>	2022	0.023	0.879		0.009	0.000
>		0.005	0.074	0.010		
>	Total	0.016	0.823		0.047	0.005
>		0.012	0.086	0.011		

(collection **Table** exported to file prop3_1.xls)

		Number of activities last year		
		None	At least 1	Not applicable
Year				
	2011	0.143	0.849	0.008
	2013	0.135	0.851	0.014
	2015	0.142	0.842	0.016
	2017	0.151	0.826	0.023
	2020	0.129	0.846	0.025
	2022	0.159	0.819	0.023
	Total	0.142	0.841	0.016

(collection **Table** exported to file prop4_1.xls)

		Gender	
		Male	Female
Year			
	2011	0.453	0.547
	2013	0.446	0.554
	2015	0.439	0.561
	2017	0.431	0.569
	2020	0.427	0.573
	2022	0.414	0.586
	Total	0.439	0.561

(collection **Table** exported to file prop5_1.xls)

Number of observations = 109,546

> Qyear		sphus	age	yedu	thinc	gender	househ~d	cjs	covid
	sphus	1.0000							
	age	0.2221	1.0000						
	yedu	-0.1931	-0.1569	1.0000					
	thinc	-0.2479	-0.1647	0.1812	1.0000				
	gender	0.0536	0.0281	-0.0894	-0.1305	1.0000			
	household	-0.1044	-0.2414	0.0614	0.3520	-0.2461	1.0000		
	cjs	-0.0111	-0.1741	-0.1169	0.0013	0.1770	0.0585	1.0000	
	covid	0.0119	0.2248	0.0222	0.0538	0.0173	-0.0620	-0.0558	1.0000

```

>
>   Qyear |    0.0239    0.3517    0.0380   -0.1173    0.0240   -0.0946   -0.1354    0.5356
> 1.0000 |
>   id    |    0.0509    0.0269   -0.0709   -0.0034   -0.0109    0.0596    0.0571    0.0131
> 0.0032 |
>   gali |    0.5331    0.2056   -0.0826   -0.1510    0.0651   -0.1078   -0.0320    0.0351
> 0.0425 |
>   rhfo |    0.1600    0.2026   -0.0138   -0.3577    0.0819   -0.1882   -0.0575   -0.0086
> 0.2083 |
>   ghfo |   -0.0817   -0.1935    0.0543   -0.0897   -0.0555    0.1440    0.0254   -0.1654
> -0.1496 |
>   lifesat |   -0.3530   -0.0323    0.0676    0.1901   -0.0471    0.1276   -0.0073    0.0244
> 0.0285 |
>   naly |   -0.1662   -0.0620    0.2503    0.1587    0.0134   -0.0167   -0.2254   -0.0098
> 0.0064 |
>   country |    0.0258   -0.0169    0.1646   -0.1998    0.0167   -0.0215   -0.1014   -0.0150
> 0.0139

```

	id	gali	rhfo	ghfo	lifesat	nal	country
id	1.0000						
gali	-0.0220	1.0000					
rhfo	-0.1021	0.1782	1.0000				
ghfo	-0.0457	-0.0535	0.2268	1.0000			
lifesat	-0.0697	-0.2261	-0.0754	0.0513	1.0000		
nal	-0.1513	-0.0710	0.0046	0.0448	0.1246	1.0000	
country	-0.0003	0.0239	0.0686	0.0168	-0.1010	0.0388	1.0000

Click to Open File: [stats_1.doc](#)

(File sk_1.doc already exists, option **append** was assumed)

Skewness and kurtosis tests for normality

Variable	Obs	Pr(skewness)	Pr(kurtosis)	Joint test	
				Adj chi2(2)	Prob>chi2
age	109,546	0.0000	0.0000	.	.

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Skewness and kurtosis tests for normality

Variable	Obs	Pr(skewness)	Pr(kurtosis)	Joint test	
				Adj chi2(2)	Prob>chi2
yedu	109,546	0.0000	0.0000	106.19	0.0000

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Skewness and kurtosis tests for normality

Variable	Obs	Pr(skewness)	Pr(kurtosis)	Joint test	
				Adj chi2(2)	Prob>chi2
thinc	109,546	0.0000	0.0000	.	.

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Skewness and kurtosis tests for normality

Variable	Obs	Pr(skewness)	Pr(kurtosis)	Joint test	
				Adj chi2(2)	Prob>chi2
lifesat	109,546	0.0000	0.0000	.	.

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(obs=109,546)

	sphus	age	yedu	thinc	lifesat
sphus	1.0000				
age	0.2220	1.0000			
yedu	-0.1868	-0.1474	1.0000		
thinc	-0.2464	-0.1715	0.1713	1.0000	
lifesat	-0.3659	-0.0458	0.0758	0.1867	1.0000

Click to Open File: [stats_1.doc](#)

	sphus	age	yedu	thinc	lifesat
sphus	1.0000				
age	0.2220*	1.0000			
yedu	-0.1868*	-0.1474*	1.0000		
thinc	-0.2464*	-0.1715*	0.1713*	1.0000	
lifesat	-0.3659*	-0.0458*	0.0758*	0.1867*	1.0000

household	Freq.	Percent	Cum.
Single person responding	37,717	34.43	34.43
Couple, both responding	62,737	57.27	91.70
Multiple, at least 1 responding	9,092	8.30	100.00
Total	109,546	100.00	

0 1 2

covid	Freq.	Percent	Cum.
0	98,339	89.77	89.77
1	11,207	10.23	100.00
Total	109,546	100.00	

0 1

Current job situation	Freq.	Percent	Cum.
Not applicable	1,763	1.61	1.61
Retired	90,106	82.25	83.86
Employed or self-employed	5,183	4.73	88.59
Unemployed	521	0.48	89.07
Permanently sick or disabled	1,337	1.22	90.29
Homemaker	9,436	8.61	98.90
Other	1,200	1.10	100.00
Total	109,546	100.00	

0 1 2 3 4 5 6

Limitation with activities	Freq.	Percent	Cum.
Not limited	52,191	47.64	47.64
Limited	57,355	52.36	100.00
Total	109,546	100.00	

0 1

Received help from others (how many)	Freq.	Percent	Cum.
None	76,427	69.77	69.77
At least 1	24,842	22.68	92.44
Not applicable	8,277	7.56	100.00
Total	109,546	100.00	

0 1 2

Given help to others (how many)	Freq.	Percent	Cum.
None	65,047	59.38	59.38
At least 1	20,687	18.88	78.26
Not applicable	23,812	21.74	100.00
Total	109,546	100.00	

0 1 2

Number of activities last year	Freq.	Percent	Cum.
None	15,604	14.24	14.24
At least 1	92,179	84.15	98.39
Not applicable	1,763	1.61	100.00
Total	109,546	100.00	

0 1 2

file **violinplot_v6_1.tif** saved as TIFF format

Variable name	Storage type	Display format	Value label	Variable label
plot11	float	%9.0g		meanPlot, country == Austria
plot12	float	%9.0g		meanPlot, country == Germany
plot13	float	%9.0g		meanPlot, country == Sweden
plot14	float	%9.0g		meanPlot, country == Netherlands
plot15	float	%9.0g		meanPlot, country == Spain
plot16	float	%9.0g		meanPlot, country == Italy
plot17	float	%9.0g		meanPlot, country == France
plot18	float	%9.0g		meanPlot, country == Denmark
plot20	float	%9.0g		meanPlot, country == Switzerland
plot23	float	%9.0g		meanPlot, country == Belgium
plot28	float	%9.0g		meanPlot, country == Czech Republic
plot34	float	%9.0g		meanPlot, country == Slovenia

file **meanplot_1.tif** saved as TIFF format(File transition_1.doc already exists, option **append** was assumed)

Self-perceived health - US scale	Self-perceived health - US scale					Total
	1	2	3	4	5	
1	36.65	33.41	22.72	6.08	1.14	100.00
2	10.33	39.81	38.69	9.25	1.91	100.00
3	2.67	13.07	56.22	23.83	4.20	100.00
4	0.77	3.42	28.71	52.26	14.84	100.00
5	0.27	1.29	10.28	34.33	53.83	100.00
Total	5.38	15.09	39.32	29.10	11.11	100.00

Click to Open File: [transition_1.doc](#)note: **2.gender** omitted because of collinearity.note: **2.naly** omitted because of collinearity.note: **2.rhfo#0b.gh**to identifies no observations in the sample.note: **2.rhfo#1.gh**to identifies no observations in the sample.note: **2.rhfo#2.gh**to omitted because of collinearity.note: **11.country** omitted because of collinearity.note: **12.country** omitted because of collinearity.note: **14.country** omitted because of collinearity.note: **15.country** omitted because of collinearity.note: **16.country** omitted because of collinearity.note: **17.country** omitted because of collinearity.note: **18.country** omitted because of collinearity.note: **20.country** omitted because of collinearity.note: **23.country** omitted because of collinearity.note: **28.country** omitted because of collinearity.note: **34.country** omitted because of collinearity.Fixed-effects (within) regression
Group variable: **id**Number of obs = 109,546
Number of groups = 29,995

R-squared:

Within = 0.1176
Between = 0.4212
Overall = 0.3207

Obs per group:

min = 1
avg = 3.7
max = 6

corr(u_i, Xb) = 0.2972

F(25, 79526) = 423.90
Prob > F = 0.0000

[95% con		P> t	t	Std. err.	Coefficient	sphus	> f. interval]
-.0592142		0.000	-6.04	.0073981	-.0447139	age	> -.0302137
.0003728		0.000	9.46	.0000497	.0004702	c.age#c.age	> .0005677
-.0167595		0.614	-0.50	.0068015	-.0034286	yedu	> .0099022
-.0004381		0.602	0.52	.0003045	.0001587	c.yedu#c.yedu	> .0007555
-.0558126		0.000	-6.73	.0064234	-.0432228	thinc	> -.0306329
.005145		0.000	3.75	.0028758	.0107815	c.thinc#c.thinc	> .0164181
-.0221884		0.680	-0.41	.0093535	0 (omitted)	gender Female 1.covid	> .014477
.051533		0.000	6.11	.0124211	.0758783	household Couple, both responding	> .1002237
.0670728		0.000	6.07	.0163019	.0990244	Multiple, at least 1 responding	> .130976
-.0225207		0.243	1.17	.0283918	.0331271	cjs Not applicable	> .0887749
.0200112		0.001	3.35	.0144175	.0482694	Retired	> .0765276
-.0645402		0.739	0.33	.0396618	.0131966	Unemployed	> .0909334
.1120382		0.000	5.87	.0286464	.168185	Permanently sick or disabled	> .2243317
.0090191		0.015	2.42	.0194453	.0471318	Homemaker	> .0852445
-.0474557		0.803	0.25	.0277423	.006919	Other	> .0612937
.4155668		0.000	71.70	.0059589	.4272461	gali Limited	> .4389255
-.0478491		0.000	-4.22	.007739	-.0326808	lifesat	> -.0175125
-.0034363		0.000	-4.28	.0005509	-.0023564	c.lifesat#c.lifesat	> -.0012766
-.0785402		0.000	-6.94	.0088267	-.0612399	naly At least 1	> -.0439395
0 (omitted)					0	Not applicable	
.0703572		0.000	11.11	.0076913	.085432	rhfo At least 1	> .1005069
-.0317982		0.792	-0.26	.0143016	-.0037672	Not applicable	> .0242638
-.0399518		0.001	-3.19	.0077534	-.0247552	ghfo At least 1	> -.0095586

>	.0211875	Not applicable	.0038869	.0088269	0.44	0.660	-.0134137
		rhfo#gheto					
>	.0058586	At least 1#At least 1	-.0207308	.0135661	-1.53	0.126	-.0473202
>	-.0226207	At least 1#Not applicable	-.0614479	.0198099	-3.10	0.002	-.1002752
		Not applicable#None	0	(empty)			
		Not applicable#At least 1	0	(empty)			
		Not applicable#Not applicable	0	(omitted)			
		country					
		Austria	0	(omitted)			
		Germany	0	(omitted)			
		Netherlands	0	(omitted)			
		Spain	0	(omitted)			
		Italy	0	(omitted)			
		France	0	(omitted)			
		Denmark	0	(omitted)			
		Switzerland	0	(omitted)			
		Belgium	0	(omitted)			
		Czech Republic	0	(omitted)			
		Slovenia	0	(omitted)			
>	4.62027	_cons	4.077612	.276867	14.73	0.000	3.534955
		sigma_u	.7279782				
		sigma_e	.64115646				
		rho	.5631596			(fraction of variance due to u_i)	

F test that all u_i=0: F(29994, 79526) = 3.09 Prob > F = 0.0000

results_impl_fe_1.xls

dir : seeout

note: 2.naly omitted because of collinearity.

note: 2.rhfo#0.gheto identifies no observations in the sample.

note: 2.rhfo#1.gheto identifies no observations in the sample.

note: 2.rhfo#2.gheto omitted because of collinearity.

Random-effects GLS regression

Group variable: id

Number of obs = 109,546

Number of groups = 29,995

R-squared:

Within = 0.1101

Between = 0.5256

Overall = 0.3944

Obs per group:

min = 1

avg = 3.7

max = 6

corr(u_i, X) = 0 (assumed)

Wald chi2(37) = 41598.71

Prob > chi2 = 0.0000

		sphus	Coefficient	Std. err.	z	P> z	[95% con
> f. interval]							
>	.0262372	age	.0144361	.0060211	2.40	0.017	.002635
>	.0000883	c.age#c.age	9.86e-06	.00004	0.25	0.805	-.0000685
>	-.0006476	yedu	-.006081	.0027722	-2.19	0.028	-.0115144
>	-.0003309	c.yedu#c.yedu	-.0005706	.0001223	-4.66	0.000	-.0008104
		thinc	-.071884	.0057198	-12.57	0.000	-.0830947

>	- .0606734					
>	.0182705	c.thinc#c.thinc	.0132372	.0025681	5.15	0.000 .0082038
		gender				
>	.01255	Female	-.0018364	.0073401	-0.25	0.802 -.0162228
>	.002257	1.covid	-.0123357	.0074454	-1.66	0.098 -.0269284
		household				
>	.0743138	Couple, both responding	.0593069	.0076567	7.75	0.000 .0442999
>	.0945009	Multiple, at least 1 responding	.0720561	.0114516	6.29	0.000 .0496113
		cjs				
>	.2193634	Not applicable	.1720366	.0241468	7.12	0.000 .1247098
>	.1317372	Retired	.1071875	.0125256	8.56	0.000 .0826378
>	.1501879	Unemployed	.0810878	.0352558	2.30	0.021 .0119876
>	.4840199	Permanently sick or disabled	.4357654	.0246201	17.70	0.000 .387511
>	.1607518	Homemaker	.1295307	.0159294	8.13	0.000 .0983096
>	.120386	Other	.0712287	.0250807	2.84	0.005 .0220714
		gali				
>	.670891	Limited	.660683	.0052083	126.85	0.000 .650475
>	-.0463716	lifesat	-.0597828	.0068426	-8.74	0.000 -.073194
>	-.0021935	c.lifesat#c.lifesat	-.0031496	.0004878	-6.46	0.000 -.0041058
		naly				
>	-.1113582	At least 1	-.1265395	.0077457	-16.34	0.000 -.1417208
		Not applicable	0	(omitted)		
		rhfo				
>	.1438666	At least 1	.1298262	.0071636	18.12	0.000 .1157857
>	-.0234294	Not applicable	-.0481933	.0126349	-3.81	0.000 -.0729572
		ghfo				
>	-.0528378	At least 1	-.067079	.0072661	-9.23	0.000 -.0813202
>	.0210556	Not applicable	.0058563	.0077548	0.76	0.450 -.0093429
		rhfo#ghfo				
>	-.0191809	At least 1#At least 1	-.044558	.0129477	-3.44	0.001 -.0699351
>	.0127751	At least 1#Not applicable	-.0219111	.0176973	-1.24	0.216 -.0565972
		Not applicable#None	0	(empty)		
		Not applicable#At least 1	0	(empty)		
		Not applicable#Not applicable	0	(omitted)		
		country				
>	.1392644	Austria	.1047625	.0176033	5.95	0.000 .0702607
>	.3728973	Germany	.329686	.022047	14.95	0.000 .2864747
		Netherlands	.0470117	.0200865	2.34	0.019 .007643

>	.0863805						
>	.3874344	Spain		.3492685	.0194727	17.94	0.000 .3111027
>	.3874344	Italy		.2666159	.0190828	13.97	0.000 .2292143
>	.3040175	France		.25466	.0174321	14.61	0.000 .2204937
>	.2888263	Denmark		-.1068227	.0210323	-5.08	0.000 -.1480453
>	-.0656	Switzerland		-.0090821	.0190797	-0.48	0.634 -.0464777
>	.0283135	Belgium		.0847549	.0176932	4.79	0.000 .0500768
>	.119433	Czech Republic		.2801516	.0181089	15.47	0.000 .2446589
>	.3156444	Slovenia		.3089664	.0203808	15.16	0.000 .2690207
>	.3489121						
>	2.802621	_cons		2.361356	.2251394	10.49	0.000 1.920091
		sigma_u		.43889503			
		sigma_e		.64115646			
		rho		.31907489	(fraction of variance due to u_i)		

results_impl_re_1.xls

dir : seeout

(File REtest_1.doc already exists, option **append** was assumed)

Mundlak specification test

H0: Covariates are uncorrelated with unobserved panel-level effects

chi2(25) = **7374.43**

Prob > chi2 = **0.0000**

Notes: Fixed effects and correlated random effects are consistent under H0 and Ha.

Random effects are efficient under H0.

Click to Open File: [REtest_1.doc](#)

Breusch and Pagan Lagrangian multiplier test for random effects

sphus[id,t] = Xb + u[id] + e[id,t]

Estimated results:

	Var	SD = sqrt(Var)
sphus	1.060887	1.029994
e	.4110816	.6411565
u	.1926288	.438895

Test: Var(u) = 0

[chibar2\(01\)](#) = **15517.97**

Prob > chibar2 = **0.0000**

Note: the rank of the differenced variance matrix (**24**) does not equal the number of coefficients being tested (**25**); be sure this is what you expect, or there may be problems computing the test. Examine the output of your estimators for anything unexpected and possibly consider scaling your variables so that the coefficients are on a similar scale.

	Coefficients		(b-B) Difference	sqrt(diag(V_b-V_B)) Std. err.
	(b) fixedR	(B) randomR		
age	-.0447139	.0144361	-.05915	.0042987
c.age#c.age	.0004702	9.86e-06	.0004604	.0000296
yedu	-.0034286	-.006081	.0026524	.0062109
c.yedu#				
c.yedu	.0001587	-.0005706	.0007293	.0002788
thinc	-.0432228	-.071884	.0286612	.0029229
c.thinc#				
c.thinc	.0107815	.0132372	-.0024556	.0012943
1.covid	-.0038557	-.0123357	.00848	.0056616
household				
1	.0758783	.0593069	.0165715	.0097805
2	.0990244	.0720561	.0269683	.0116022
c.js				
0	.0331271	.1720366	-.1389095	.0149341
1	.0482694	.1071875	-.0589181	.0071396
3	.0131966	.0810878	-.0678912	.0181682
4	.168185	.4357654	-.2675805	.0146447
5	.0471318	.1295307	-.082399	.0111523
6	.006919	.0712287	-.0643097	.0118572
1.gali	.4272461	.660683	-.2334369	.0028952
lifesat	-.0326808	-.0597828	.0271019	.0036153
c.lifesat#				
c.lifesat	-.0023564	-.0031496	.0007932	.000256
1.naly	-.0612399	-.1265395	.0652997	.0042326
rhfo				
1	.085432	.1298262	-.0443941	.0027997
2	-.0037672	-.0481933	.0444262	.0067004
ghfo				
1	-.0247552	-.067079	.0423238	.0027054
2	.0038869	.0058563	-.0019694	.0042161
rhfo#ghfo				
1 1	-.0207308	-.044558	.0238272	.004049
1 2	-.0614479	-.0219111	-.0395369	.0089015

b = Consistent under H0 and Ha; obtained from **xtreg**.
 B = Inconsistent under Ha, efficient under H0; obtained from **xtreg**.

Test of H0: Difference in coefficients not systematic

```
chi2(24) = (b-B)'[(V_b-V_B)^(-1)](b-B)
          = 7950.30
Prob > chi2 = 0.0000
```

(V_b-V_B is not positive definite)

Click to Open File: [REtest_1.doc](#)

note: **2.gender** omitted from **xt_means** because of collinearity.
 note: **2.naly** omitted from **xt_means** because of collinearity.
 note: **10.rhfo#0b.ghfo** omitted from **xt_means** because of collinearity.
 note: **20.rhfo#0b.ghfo** omitted from **xt_means** because of collinearity.
 note: **20.rhfo#10.ghfo** omitted from **xt_means** because of collinearity.
 note: **2.rhfo#2.ghfo** omitted from **xt_means** because of collinearity.
 note: **11.country** omitted from **xt_means** because of collinearity.
 note: **12.country** omitted from **xt_means** because of collinearity.
 note: **13b.country** omitted from **xt_means** because of collinearity.
 note: **14.country** omitted from **xt_means** because of collinearity.
 note: **15.country** omitted from **xt_means** because of collinearity.
 note: **16.country** omitted from **xt_means** because of collinearity.
 note: **17.country** omitted from **xt_means** because of collinearity.
 note: **18.country** omitted from **xt_means** because of collinearity.
 note: **20.country** omitted from **xt_means** because of collinearity.
 note: **23.country** omitted from **xt_means** because of collinearity.
 note: **28.country** omitted from **xt_means** because of collinearity.
 note: **34.country** omitted from **xt_means** because of collinearity.

Correlated random-effects regression
 Group variable: **id**

Number of obs = 109,546
 Number of groups = 29,995

R-squared:
 Within = 0.1176
 Between = 0.5598
 Overall = 0.4470

Obs per group:
 min = 1
 avg = 3.7
 max = 6

corr(xit_vars*b, xt_means*y) = 0.5671

Wald chi2(37) = 8373.08
 Prob > chi2 = 0.0000

(Std. err. adjusted for 29,995 clu

> sters in id)

		sphus	Coefficient	Robust std. err.	z	P> z	[95% con
> f. interval]							
xit_vars							
>	-.0281533	age	-.0447139	.0084494	-5.29	0.000	-.0612745
>	.0005824	c.age#c.age	.0004702	.0000572	8.22	0.000	.0003581
>	.0096438	yedu	-.0034286	.0066697	-0.51	0.607	-.0165011
>	.0007377	c.yedu#c.yedu	.0001587	.0002954	0.54	0.591	-.0004203
>	-.0301661	thinc	-.0432228	.0066617	-6.49	0.000	-.0562795
>	.0165526	c.thinc#c.thinc	.0107815	.0029445	3.66	0.000	.0050105
>	-.0033346	gender Female	-.0179708	.0074676	-2.41	0.016	-.032607
>	.014317	1.covid	-.0038557	.0092719	-0.42	0.678	-.0220284
>		household					
>	.102332	Couple, both responding	.0758783	.013497	5.62	0.000	.0494247
>	.1331492	Multiple, at least 1 responding	.0990244	.0174109	5.69	0.000	.0648996
>		cjs					
>	.0989008	Not applicable	.0331271	.0335586	0.99	0.324	-.0326465
>	.0791273	Retired	.0482694	.0157441	3.07	0.002	.0174115
>	.092358	Unemployed	.0131966	.0403892	0.33	0.744	-.0659648
>	.2306179	Permanently sick or disabled	.168185	.0318541	5.28	0.000	.105752
>	.0878839	Homemaker	.0471318	.0207923	2.27	0.023	.0063797
>	.0636493	Other	.006919	.0289445	0.24	0.811	-.0498112
>	.440108	gali Limited	.4272461	.0065623	65.11	0.000	.4143842
>	-.0170471	lifesat	-.0326808	.0079766	-4.10	0.000	-.0483146
>	-.0012337	c.lifesat#c.lifesat	-.0023564	.0005728	-4.11	0.000	-.0034792
		naly					

>	-.0421238	At least 1	-.0612399	.0097533	-6.28	0.000	-.0803559
		Not applicable	0	(omitted)			
		rhfo					
>	.1009481	At least 1	.085432	.0079165	10.79	0.000	.0699159
>	.0247984	Not applicable	-.0037672	.0145745	-0.26	0.796	-.0323327
		ghfo					
>	-.0098102	At least 1	-.0247552	.0076251	-3.25	0.001	-.0397002
>	.021842	Not applicable	.0038869	.0091609	0.42	0.671	-.0140682
		rhfo#ghfo					
>	.0054536	At least 1#At least 1	-.0207308	.0133596	-1.55	0.121	-.0469151
>	-.021529	At least 1#Not applicable	-.0614479	.0203672	-3.02	0.003	-.1013669
		Not applicable#None	0	(empty)			
		Not applicable#At least 1	0	(empty)			
		Not applicable#Not applicable	0	(omitted)			
		country					
>	.0968967	Austria	.0596997	.0189784	3.15	0.002	.0225027
>	.2466054	Germany	.204516	.0214746	9.52	0.000	.1624266
>	.0419297	Netherlands	-.0009063	.0218555	-0.04	0.967	-.0437423
>	.3505603	Spain	.3094234	.0209886	14.74	0.000	.2682866
>	.2532623	Italy	.2127426	.0206737	10.29	0.000	.172223
>	.205239	France	.168117	.0189401	8.88	0.000	.130995
>	-.0542961	Denmark	-.0992594	.0229409	-4.33	0.000	-.1442227
>	.1129419	Switzerland	.0720668	.020855	3.46	0.001	.0311917
>	.0541968	Belgium	.0172372	.0188573	0.91	0.361	-.0197224
>	.1649839	Czech Republic	.1230336	.0214036	5.75	0.000	.0810833
>	.2564725	Slovenia	.2133449	.0220043	9.70	0.000	.1702173
>	1.093438	_cons	.396891	.3553876	1.12	0.264	-.2996558
xt_means							
>	.1467062	age	.1217323	.012742	9.55	0.000	.0967583
>	-.0007825	c.age#c.age	-.0009493	.0000851	-11.16	0.000	-.0011161
>	.0190708	yedu	.0048834	.0072386	0.67	0.500	-.009304
>	-.0001876	c.yedu#c.yedu	-.0008173	.0003213	-2.54	0.011	-.001447
>	-.0065376	thinc	-.0322287	.0131079	-2.46	0.014	-.0579197
>	.0031384	c.thinc#c.thinc	-.0084604	.0059179	-1.43	0.153	-.0200592

	gender					
	Female	0	(omitted)			
	1.covid	-.5334799	.0373732	-14.27	0.000	-.60673
>	-.4602298					
	household					
	Couple, both responding	.0022613	.0170289	0.13	0.894	-.0311147
>	.0356372					
	Multiple, at least 1 responding	-.005457	.0241265	-0.23	0.821	-.052744
>	.0418301					
	cjs					
	Not applicable	.2197715	.0546268	4.02	0.000	.1127048
>	.3268381					
	Retired	.1141175	.0291661	3.91	0.000	.056953
>	.171282					
	Unemployed	.0906397	.0811444	1.12	0.264	-.0684005
>	.2496799					
	Permanently sick or disabled	.4294283	.0546759	7.85	0.000	.3222654
>	.5365911					
	Homemaker	.1307292	.0345628	3.78	0.000	.0629874
>	.198471					
	Other	.0012069	.0603662	0.02	0.984	-.1171087
>	.1195225					
	gali					
	Limited	.72304	.0120412	60.05	0.000	.6994398
>	.7466403					
	lifesat	-.040804	.0150432	-2.71	0.007	-.070288
>	-.0113199					
	c.lifesat#c.lifesat	-.0025123	.0010891	-2.31	0.021	-.004647
>	-.0003776					
	naly					
	At least 1	-.0689765	.0172119	-4.01	0.000	-.1027112
>	-.0352419					
	Not applicable	0	(omitted)			
	rhfo					
	At least 1	.0973787	.0180538	5.39	0.000	.0619938
>	.1327635					
	Not applicable	-.1005713	.0420404	-2.39	0.017	-.1829689
>	-.0181736					
	ghfo					
	At least 1	-.136728	.0191156	-7.15	0.000	-.1741939
>	-.099262					
	Not applicable	-.0079566	.0193459	-0.41	0.681	-.0458739
>	.0299606					
	rhfo#ghfo					
	At least 1#At least 1	-.0943108	.0367563	-2.57	0.010	-.1663518
>	-.0222698					
	At least 1#Not applicable	.0874614	.0403676	2.17	0.030	.0083424
>	.1665805					
	Not applicable#None	0	(empty)			
	Not applicable#At least 1	0	(empty)			
	Not applicable#Not applicable	0	(omitted)			
	country					
	Austria	0	(omitted)			
	Germany	0	(omitted)			
	Netherlands	0	(omitted)			
	Spain	0	(omitted)			
	Italy	0	(omitted)			
	France	0	(omitted)			
	Denmark	0	(omitted)			
	Switzerland	0	(omitted)			
	Belgium	0	(omitted)			
	Czech Republic	0	(omitted)			
	Slovenia	0	(omitted)			

	sigma_u	.43889503	
	sigma_e	.64115646	
	rho	.31907489	(fraction of variance due to u_i)

Mundlak test (xt_means = 0): chi2(25) = 6.8e+03 Prob > chi2 = 0.0000

results_impl_cre_1.xls

dir : seeout

note: 2.naly omitted because of collinearity.
 note: 2.rhfo#0.gho identifies no observations in the sample.
 note: 2.rhfo#1.gho identifies no observations in the sample.
 note: 2.rhfo#2.gho omitted because of collinearity.
 note: Mundlak_household_Multiple omitted because of collinearity.
 note: Mundlak_covid_1 omitted because of collinearity.
 note: Mundlak_cjs_Other omitted because of collinearity.
 note: Mundlak_gali_Limited omitted because of collinearity.
 note: Mundlak_rhfo_Not omitted because of collinearity.
 note: Mundlak_gho_Not omitted because of collinearity.
 note: Mundlak_naly_At omitted because of collinearity.
 note: Mundlak_naly_Not omitted because of collinearity.
 note: Mundlak_i2 omitted because of collinearity.
 note: Mundlak_i3 omitted because of collinearity.
 note: Mundlak_i5 omitted because of collinearity.
 note: Mundlak_i6 omitted because of collinearity.
 note: Mundlak_i7 omitted because of collinearity.
 note: Mundlak_i8 omitted because of collinearity.
 note: Mundlak_i9 omitted because of collinearity.

Random-effects GLS regression

Group variable: id

Number of obs = 109,546

Number of groups = 29,995

R-squared:

Within = 0.1176

Between = 0.5598

Overall = 0.4470

Obs per group:

min = 1

avg = 3.7

max = 6

corr(u_i, X) = 0 (assumed)

Wald chi2(62) = 52812.92

Prob > chi2 = 0.0000

(Std. err. adjusted for 29,995 clu

> sters in id)

	sphus	Coefficient	Robust std. err.	z	P> z	[95% con
> f. interval]						
> -.0281533	age	-.0447139	.0084494	-5.29	0.000	-.0612745
> .0005824	c.age#c.age	.0004702	.0000572	8.22	0.000	.0003581
> .0096438	yedu	-.0034286	.0066697	-0.51	0.607	-.0165011
> .0007377	c.yedu#c.yedu	.0001587	.0002954	0.54	0.591	-.0004203
> -.0301661	thinc	-.0432228	.0066617	-6.49	0.000	-.0562795
> .0165526	c.thinc#c.thinc	.0107815	.0029445	3.66	0.000	.0050105
> -.0033346	gender Female	-.0179708	.0074676	-2.41	0.016	-.032607
	1.covid	-.0038557	.0092719	-0.42	0.678	-.0220284

>	.014317					
		household				
		Couple, both responding	.0758783	.013497	5.62	0.000 .0494247
>	.102332					
		Multiple, at least 1 responding	.0990244	.0174109	5.69	0.000 .0648996
>	.1331492					
		cjs				
		Not applicable	.0331271	.0335586	0.99	0.324 -.0326465
>	.0989007					
		Retired	.0482694	.0157441	3.07	0.002 .0174115
>	.0791273					
		Unemployed	.0131966	.0403892	0.33	0.744 -.0659648
>	.092358					
		Permanently sick or disabled	.168185	.0318541	5.28	0.000 .105752
>	.2306179					
		Homemaker	.0471318	.0207923	2.27	0.023 .0063797
>	.0878839					
		Other	.006919	.0289445	0.24	0.811 -.0498112
>	.0636493					
		gali				
		Limited	.4272461	.0065623	65.11	0.000 .4143842
>	.440108					
		lifesat	-.0326808	.0079766	-4.10	0.000 -.0483146
>	-.0170471					
		c.lifesat#c.lifesat	-.0023564	.0005728	-4.11	0.000 -.0034792
>	-.0012337					
		naly				
		At least 1	-.0612399	.0097533	-6.28	0.000 -.0803559
>	-.0421238					
		Not applicable	0	(omitted)		
		rhfo				
		At least 1	.085432	.0079165	10.79	0.000 .0699159
>	.1009481					
		Not applicable	-.0037672	.0145745	-0.26	0.796 -.0323327
>	.0247984					
		ghfo				
		At least 1	-.0247552	.0076251	-3.25	0.001 -.0397002
>	-.0098102					
		Not applicable	.0038869	.0091609	0.42	0.671 -.0140682
>	.021842					
		rhfo#ghfo				
		At least 1#At least 1	-.0207308	.0133596	-1.55	0.121 -.0469151
>	.0054536					
		At least 1#Not applicable	-.0614479	.0203672	-3.02	0.003 -.1013669
>	-.021529					
		Not applicable#None	0	(empty)		
		Not applicable#At least 1	0	(empty)		
		Not applicable#Not applicable	0	(omitted)		
		country				
		Austria	.0596997	.0189784	3.15	0.002 .0225027
>	.0968968					
		Germany	.204516	.0214746	9.52	0.000 .1624266
>	.2466054					
		Netherlands	-.0009064	.0218555	-0.04	0.967 -.0437424
>	.0419296					
		Spain	.3094234	.0209886	14.74	0.000 .2682866
>	.3505603					
		Italy	.2127426	.0206737	10.29	0.000 .172223
>	.2532623					
		France	.168117	.0189401	8.88	0.000 .130995
>	.205239					
		Denmark	-.0992594	.0229409	-4.33	0.000 -.1442227
>	-.0542961					

>	.1129419	Switzerland	.0720668	.020855	3.46	0.001	.0311917
>	.0541967	Belgium	.0172371	.0188573	0.91	0.361	-.0197225
>	.1649839	Czech Republic	.1230336	.0214036	5.75	0.000	.0810833
>	.2564724	Slovenia	.2133448	.0220043	9.70	0.000	.1702172
>	.1467062	Mundlak_age	.1217323	.012742	9.55	0.000	.0967583
>	-.0065376	Mundlak_thinc	-.0322287	.0131079	-2.46	0.014	-.0579197
>	-.0113199	Mundlak_lifesat	-.0408039	.0150432	-2.71	0.007	-.070288
>	.0190708	Mundlak_yedu	.0048834	.0072386	0.67	0.500	-.009304
>	-.0007825	Mundlak_age2	-.0009493	.0000851	-11.16	0.000	-.0011161
>	.0031384	Mundlak_thinc2	-.0084604	.0059179	-1.43	0.153	-.0200592
>	-.0003776	Mundlak_lifesat2	-.0025123	.0010891	-2.31	0.021	-.004647
>	-.0001876	Mundlak_yedu2	-.0008173	.0003213	-2.54	0.011	-.001447
>	.052744	Mundlak_household_Single	.005457	.0241265	0.23	0.821	-.0418301
>	.0486299	Mundlak_household_Couple	.0077182	.0208737	0.37	0.712	-.0331935
>		Mundlak_household_Multiple	0	(omitted)			
>	.6067299	Mundlak_covid_0	.5334798	.0373732	14.27	0.000	.4602298
>		Mundlak_covid_1	0	(omitted)			
>	.4220337	Mundlak_cjs_Not	.2875411	.06862	4.19	0.000	.1530484
>	.2175759	Mundlak_cjs_Retired	.1129106	.0534017	2.11	0.034	.0082452
>	.1171086	Mundlak_cjs_Employed	-.0012069	.0603662	-0.02	0.984	-.1195225
>	.2745906	Mundlak_cjs_Unemployed	.0894327	.09447	0.95	0.344	-.0957251
>	.5664647	Mundlak_cjs_Permanently	.4282213	.0705336	6.07	0.000	.2899779
>	.2390737	Mundlak_cjs_Homemaker	.1295223	.0558946	2.32	0.020	.0199709
>		Mundlak_cjs_Other	0	(omitted)			
>	-.6994398	Mundlak_gali_Not	-.72304	.0120412	-60.05	0.000	-.7466403
>		Mundlak_gali_Limited	0	(omitted)			
>	.1829687	Mundlak_rhfo_None	.1005711	.0420404	2.39	0.017	.0181734
>	.3838961	Mundlak_rhfo_At	.2854112	.0502484	5.68	0.000	.1869262
>		Mundlak_rhfo_Not	0	(omitted)			
>	-.0062848	Mundlak_ghfo_None	-.0795048	.0373578	-2.13	0.033	-.1527248
>	-.226576	Mundlak_ghfo_At	-.3105435	.0428414	-7.25	0.000	-.3945111
>		Mundlak_ghfo_Not	0	(omitted)			
>	.1027112	Mundlak_naly_None	.0689765	.0172119	4.01	0.000	.0352419
>		Mundlak_naly_At	0	(omitted)			
>		Mundlak_naly_Not	0	(omitted)			
>	.1665804	Mundlak_i1	.0874614	.0403676	2.17	0.030	.0083424
>		Mundlak_i2	0	(omitted)			
>		Mundlak_i3	0	(omitted)			
>	.2759927	Mundlak_i4	.1817722	.0480726	3.78	0.000	.0875517
		Mundlak_i5	0	(omitted)			
		Mundlak_i6	0	(omitted)			
		Mundlak_i7	0	(omitted)			

	Mundlak_i8	0	(omitted)			
	Mundlak_i9	0	(omitted)			
>	_cons	.4046973	.3726741	1.09	0.278	-.3257305
	sigma_u	.43889503				
	sigma_e	.64115646				
	rho	.31907489	(fraction of variance due to u_i)			

results_impl_Mundlak_1.xls

dir : seeout

(File Mundlak_1.doc already exists, option **append** was assumed)

```

( 1) Mundlak_age = 0
( 2) Mundlak_thinc = 0
( 3) Mundlak_lifesat = 0
( 4) Mundlak_yedu = 0
( 5) Mundlak_age2 = 0
( 6) Mundlak_thinc2 = 0
( 7) Mundlak_lifesat2 = 0
( 8) Mundlak_yedu2 = 0
( 9) Mundlak_household_Single = 0
(10) Mundlak_household_Couple = 0
(11) o.Mundlak_household_Multiple = 0
(12) Mundlak_covid_0 = 0
(13) o.Mundlak_covid_1 = 0
(14) Mundlak_cjs_Not = 0
(15) Mundlak_cjs_Retired = 0
(16) Mundlak_cjs_Employed = 0
(17) Mundlak_cjs_Unemployed = 0
(18) Mundlak_cjs_Permanently = 0
(19) Mundlak_cjs_Homemaker = 0
(20) o.Mundlak_cjs_Other = 0
(21) Mundlak_gali_Not = 0
(22) o.Mundlak_gali_Limited = 0
(23) Mundlak_rhfo_None = 0
(24) Mundlak_rhfo_At = 0
(25) o.Mundlak_rhfo_Not = 0
(26) Mundlak_ghfo_None = 0
(27) Mundlak_ghfo_At = 0
(28) o.Mundlak_ghfo_Not = 0
(29) Mundlak_naly_None = 0
(30) o.Mundlak_naly_At = 0
(31) o.Mundlak_naly_Not = 0
(32) Mundlak_i1 = 0
(33) o.Mundlak_i2 = 0
(34) o.Mundlak_i3 = 0
(35) Mundlak_i4 = 0
(36) o.Mundlak_i5 = 0
(37) o.Mundlak_i6 = 0
(38) o.Mundlak_i7 = 0
(39) o.Mundlak_i8 = 0
(40) o.Mundlak_i9 = 0
Constraint 11 dropped
Constraint 13 dropped
Constraint 20 dropped
Constraint 22 dropped
Constraint 25 dropped
Constraint 28 dropped
Constraint 30 dropped
Constraint 31 dropped
Constraint 33 dropped
Constraint 34 dropped
Constraint 36 dropped
Constraint 37 dropped
Constraint 38 dropped
Constraint 39 dropped
Constraint 40 dropped

```

```

      chi2( 25) = 6817.66
      Prob > chi2 = 0.0000
Click to Open File: Mundlak\_1.doc
note: group() not specified; assuming group(id) from panel identifier

```

```

note: 2.naly omitted because of collinearity.
note: 2.rhfo#0.gho identifies no observations in the sample.
note: 2.rhfo#1.gho identifies no observations in the sample.
note: 2.rhfo#2.gho omitted because of collinearity.
note: multiple positive outcomes within groups encountered.
note: 2.gender omitted because of no within-group variance.
note: 11.country omitted because of no within-group variance.
note: 12.country omitted because of no within-group variance.
note: 14.country omitted because of no within-group variance.
note: 15.country omitted because of no within-group variance.
note: 16.country omitted because of no within-group variance.
note: 17.country omitted because of no within-group variance.
note: 18.country omitted because of no within-group variance.
note: 20.country omitted because of no within-group variance.
note: 23.country omitted because of no within-group variance.
note: 28.country omitted because of no within-group variance.
note: 34.country omitted because of no within-group variance.

```

```

Iteration 0: Log conditional likelihood = -99532938
Iteration 1: Log conditional likelihood = -95437552
Iteration 2: Log conditional likelihood = -95431617
Iteration 3: Log conditional likelihood = -95431616

```

Fixed-effects ordered logistic regression

```

                                N. of obs. (inc. copies) = 130636
                                N. of observations      = 86544
                                N. of panel units       = 19389
                                Wald chi2(25)          = 2777.17
                                Prob > chi2            = 0.0000
                                Pseudo R2              = 0.1295
Log conditional likelihood = -95431616 (Std. err. adjusted for 19,389 clu
> sters in id)

```

	sphus	Odds Ratio	Robust std. err.	z	P> z	[95% con
> f. interval]						
> 1.09857	age	.9932806	.0510592	-0.13	0.896	.8980826
> 1.001246	c.age#c.age	1.000568	.0003458	1.64	0.101	.9998903
> 1.060187	yedu	.9771901	.0406437	-0.55	0.579	.9006903
> 1.005041	c.yedu#c.yedu	1.001496	.0018058	0.83	0.407	.9979628
> .9793997	thinc	.9077158	.0352017	-2.50	0.013	.8412785
> 1.06602	c.thinc#c.thinc	1.028461	.0188214	1.53	0.125	.9922254
> 1.157237	gender Female 1.covid	1 (omitted) 1.041761	.0558748	0.76	0.446	.9378085
> 1.46517	household Couple, both responding	1.269077	.0930339	3.25	0.001	1.099228
Multiple, at least 1 responding		1.273807	.1296037	2.38	0.017	1.043513

>	1.554924						
		cjs					
		Not applicable	.8081495	.1675466	-1.03	0.304	.5382939
>	1.213288						
		Retired	1.028603	.0961054	0.30	0.763	.856481
>	1.235316						
		Unemployed	1.031488	.203754	0.16	0.875	.7003621
>	1.519167						
		Permanently sick or disabled	1.539228	.2468479	2.69	0.007	1.124073
>	2.107711						
		Homemaker	.9450132	.1105722	-0.48	0.629	.7513498
>	1.188594						
		Other	.8292247	.1188293	-1.31	0.191	.6261719
>	1.098123						
		gali					
		Limited	4.121994	.1496094	39.02	0.000	3.838952
>	4.425905						
		lifesat	.8584021	.0435493	-3.01	0.003	.7771534
>	.9481451						
		c.lifesat#c.lifesat	.9952133	.003509	-1.36	0.174	.9883595
>	1.002115						
		naly					
		At least 1	.8629428	.0386372	-3.29	0.001	.7904429
>	.9420925						
		Not applicable	1	(omitted)			
		rhfo					
		At least 1	1.223051	.0589093	4.18	0.000	1.112874
>	1.344137						
		Not applicable	1.039809	.0835135	0.49	0.627	.8883584
>	1.217079						
		ghfo					
		At least 1	.903431	.0431497	-2.13	0.033	.8226969
>	.9920879						
		Not applicable	1.010128	.0484769	0.21	0.834	.9194468
>	1.109753						
		rhfo#ghfo					
		At least 1#At least 1	.9564735	.0818309	-0.52	0.603	.8088138
>	1.13109						
		At least 1#Not applicable	1.025882	.1179509	0.22	0.824	.8188991
>	1.285181						
		Not applicable#None	1	(empty)			
		Not applicable#At least 1	1	(empty)			
		Not applicable#Not applicable	1	(omitted)			
		country					
		Austria	1	(omitted)			
		Germany	1	(omitted)			
		Netherlands	1	(omitted)			
		Spain	1	(omitted)			
		Italy	1	(omitted)			
		France	1	(omitted)			
		Denmark	1	(omitted)			
		Switzerland	1	(omitted)			
		Belgium	1	(omitted)			
		Czech Republic	1	(omitted)			
		Slovenia	1	(omitted)			

results_impl_feo_1.xls

dir : seeout

note: 2.naly omitted because of collinearity.

note: 2.rhfo#0.ghfo identifies no observations in the sample.

note: 2.rhfo#1.ghfo identifies no observations in the sample.

note: 2.rhfo#2.ghfo omitted because of collinearity.

note: Mundlak_household_Multiple omitted because of collinearity.

note: **Mundlak_covid_1** omitted because of collinearity.
 note: **Mundlak_cjs_Other** omitted because of collinearity.
 note: **Mundlak_gali_Limited** omitted because of collinearity.
 note: **Mundlak_rhfo_Not** omitted because of collinearity.
 note: **Mundlak_ghfo_Not** omitted because of collinearity.
 note: **Mundlak_naly_At** omitted because of collinearity.
 note: **Mundlak_naly_Not** omitted because of collinearity.
 note: **Mundlak_i2** omitted because of collinearity.
 note: **Mundlak_i3** omitted because of collinearity.
 note: **Mundlak_i5** omitted because of collinearity.
 note: **Mundlak_i6** omitted because of collinearity.
 note: **Mundlak_i7** omitted because of collinearity.
 note: **Mundlak_i8** omitted because of collinearity.
 note: **Mundlak_i9** omitted because of collinearity.

Fitting comparison model:

Iteration 0: Log likelihood = **-3.899e+08**
 Iteration 1: Log likelihood = **-2.821e+08**
 Iteration 2: Log likelihood = **-2.702e+08**
 Iteration 3: Log likelihood = **-2.701e+08**
 Iteration 4: Log likelihood = **-2.701e+08**
 Iteration 5: Log likelihood = **-2.701e+08**

Refining starting values:

Grid node 0: Log likelihood = **-2.620e+08**

Fitting full model:

Iteration 0: Log pseudolikelihood = **-2.620e+08**
 Iteration 1: Log pseudolikelihood = **-2.581e+08**
 Iteration 2: Log pseudolikelihood = **-2.581e+08**
 Iteration 3: Log pseudolikelihood = **-2.581e+08**

Random-effects ordered logistic regression
 Group variable: **id**

Number of obs = **109,528**
 Number of groups = **29,989**

Random effects u_i ~ Gaussian

Obs per group:
 min = **1**
 avg = **3.7**
 max = **6**

Integration method: **mvaghermite**

Integration pts. = **12**

Log pseudolikelihood = **-2.581e+08**

Wald chi2(62) = **11837.18**
 Prob > chi2 = **0.0000**

(Std. err. adjusted for **29,989** cl

> usters in **id**)

	sphus	Odds ratio	Robust std. err.	z	P> z	[95% co
> n						
> f. interval]						
> 2	age	.9534153	.0440865	-1.03	0.302	.870807
> 1.04386						
> 2	c.age#c.age	1.000833	.0003132	2.66	0.008	1.0002
> 1.001447						
> 8	yedu	.9913103	.0344735	-0.25	0.802	.925994
> 1.061233						
> 4	c.yedu#c.yedu	1.000849	.0015084	0.56	0.573	.997897

>	1.00381						
>		thinc	.897065	.0316582	-3.08	0.002	.837113
> 5	.9613102						
>		c.thinc#c.thinc	1.034407	.016973	2.06	0.039	1.0016
> 7	1.068214						
>		gender					
>		Female	1.00151	.0398297	0.04	0.970	.926410
> 4	1.082698						
>		1.covid	1.013647	.0493118	0.28	0.781	.921462
> 5	1.115054						
>		household					
>		Couple, both responding	1.206331	.0869636	2.60	0.009	1.04737
> 9	1.389406						
>		Multiple, at least 1 responding	1.22209	.1198608	2.04	0.041	1.00836
> 7	1.481112						
>		cjs					
>		Not applicable	.7608837	.1747077	-1.19	0.234	.485146
> 1	1.193339						
>		Retired	1.059927	.0901797	0.68	0.494	.897128
> 7	1.252267						
>		Unemployed	1.005843	.1890796	0.03	0.975	.695857
> 6	1.453919						
>		Permanently sick or disabled	1.857205	.3076467	3.74	0.000	1.34232
> 6	2.569577						
>		Homemaker	.9866074	.1078616	-0.12	0.902	.796317
> 2	1.22237						
>		Other	.8861326	.1224557	-0.87	0.382	.675880
> 9	1.161789						
>		gali					
>		Limited	3.90337	.1309632	40.59	0.000	3.65494
> 4	4.168681						
>		lifesat	.8068075	.0401592	-4.31	0.000	.731814
> 4	.8894854						
>		c.lifesat#c.lifesat	.9990885	.0034638	-0.26	0.793	.992322
> 7	1.005901						
>		naly					
>		At least 1	.812533	.0373814	-4.51	0.000	.742472
> 8	.8892041						
>		Not applicable	1	(omitted)			
>		rhfo					
>		At least 1	1.224194	.0562436	4.40	0.000	1.11877
> 7	1.339545						
>		Not applicable	1.05346	.0798069	0.69	0.492	.908100
> 3	1.222088						
>							

> 6		gheto					
>	1.006737	At least 1	.9281848	.0384726	-1.80	0.072	.855761
> 5		Not applicable					
>	1.078022		.9863679	.0447162	-0.30	0.762	.902506
> 9		rhfo#gheto					
>	1.097564	At least 1#At least 1	.9420066	.073457	-0.77	0.444	.808495
> 1		At least 1#Not applicable					
>	1.24369		1.011505	.1066452	0.11	0.914	.822667
		Not applicable#None	1	(empty)			
		Not applicable#At least 1	1	(empty)			
		Not applicable#Not applicable	1	(omitted)			
> 7		country					
>	1.345361	Austria	1.183349	.0774707	2.57	0.010	1.04084
> 6		Germany					
>	2.156165		1.866365	.1374454	8.47	0.000	1.61551
> 5		Netherlands					
>	1.218385		1.054242	.0778346	0.72	0.474	.912212
> 9		Spain					
>	3.80914		3.260562	.2586933	14.90	0.000	2.79098
> 9		Italy					
>	2.621397		2.261578	.1703652	10.83	0.000	1.95114
> 6		France					
>	2.026801		1.780748	.1175907	8.74	0.000	1.56456
> 1		Denmark					
>	.8627254		.7424433	.0568775	-3.89	0.000	.638931
> 5		Switzerland					
>	1.417767		1.225013	.091335	2.72	0.006	1.05846
> 9		Belgium					
>	1.218355		1.074418	.0689188	1.12	0.263	.947485
> 7		Czech Republic					
>	1.654935		1.391808	.122963	3.74	0.000	1.17051
> 1		Slovenia					
>	2.612463		2.198101	.1936827	8.94	0.000	1.84946
> 8		Mundlak_age					
>	1.655894		1.442989	.1013235	5.22	0.000	1.25745
> 9		Mundlak_thinc					
>	1.028716		.900283	.0612558	-1.54	0.123	.787884
> 8		Mundlak_lifesat					
>	.8968874		.7448809	.0705772	-3.11	0.002	.618636
> 3		Mundlak_yedu					
>	1.071238		.9959582	.0370266	-0.11	0.913	.925968
> 7		Mundlak_age2					
>	.9980455		.9971258	.000469	-6.12	0.000	.99620
> 1		Mundlak_thinc2					
>	1.031026		.968192	.0310615	-1.01	0.314	.909187

> 2	Mundlak_lifesat2	1.002924	.0066307	0.44	0.659	.990012
>	1.016005					
> 9	Mundlak_yedu2	.9976596	.0016004	-1.46	0.144	.994527
>	1.000801					
> 3	Mundlak_household_Single	.8528608	.1107428	-1.23	0.220	.661227
>	1.100033					
> 5	Mundlak_household_Couple	.9014868	.0948547	-0.99	0.324	.733492
>	1.107957					
> 2	Mundlak_household_Multiple	1	(omitted)			
>	Mundlak_covid_0	5.085378	.9399787	8.80	0.000	3.53987
>	7.30565					
> 7	Mundlak_covid_1	1	(omitted)			
>	Mundlak_cjs_Not	4.334491	1.899668	3.35	0.001	1.83604
>	10.23275					
> 1	Mundlak_cjs_Retired	1.985488	.4929656	2.76	0.006	1.22047
>	3.230036					
> 3	Mundlak_cjs_Employed	1.348218	.3889614	1.04	0.300	.765928
>	2.373188					
> 2	Mundlak_cjs_Unemployed	.9697124	.4281043	-0.07	0.944	.408186
>	2.303709					
> 2	Mundlak_cjs_Permanently	4.968038	1.627008	4.89	0.000	2.61469
>	9.439507					
> 4	Mundlak_cjs_Homemaker	1.912527	.4977618	2.49	0.013	1.1483
>	3.185259					
> 4	Mundlak_cjs_Other	1	(omitted)			
>	Mundlak_gali_Not	.1072857	.0069627	-34.40	0.000	.094471
>	.1218381					
> 8	Mundlak_gali_Limited	1	(omitted)			
>	Mundlak_rhfo_None	1.677993	.3564542	2.44	0.015	1.10654
>	2.544544					
> 1	Mundlak_rhfo_At	3.320516	.8757203	4.55	0.000	1.98024
>	5.567921					
> 8	Mundlak_rhfo_Not	1	(omitted)			
>	Mundlak_ghfo_None	.922567	.1913618	-0.39	0.698	.614382
>	1.385342					
> 9	Mundlak_ghfo_At	.3549677	.0906051	-4.06	0.000	.215238
>	.5854059					
> 3	Mundlak_ghfo_Not	1	(omitted)			
>	Mundlak_naly_None	1.08142	.0889774	0.95	0.341	.92036
>	1.270662					
> 3	Mundlak_naly_At	1	(omitted)			
>	Mundlak_naly_Not	1	(omitted)			
>	Mundlak_i1	1.027447	.2287124	0.12	0.903	.664170
>	1.589421					
> 4	Mundlak_i2	1	(omitted)			
>	Mundlak_i3	1	(omitted)			
>	Mundlak_i4	1.870288	.5154184	2.27	0.023	1.08976
>	3.209849					
>	Mundlak_i5	1	(omitted)			
>	Mundlak_i6	1	(omitted)			
>	Mundlak_i7	1	(omitted)			
>	Mundlak_i8	1	(omitted)			
>	Mundlak_i9	1	(omitted)			

		/cut1	5.913084	2.04678	1.90146
> 8	9.9247				
		/cut2	8.078348	2.047755	4.06482
> 1	12.09187				
		/cut3	11.43476	2.047698	7.42134
> 3	15.44817				
		/cut4	14.6773	2.047822	10.6636
> 4	18.69095				
		/sigma2_u	1.66296	.0552044	1.55820
> 7	1.774757				

Note: Estimates are transformed only in the first equation to odds ratios.

results_impl_xto_1.xls

dir : seeout

(File xto_1.doc already exists, option **append** was assumed)

```

( 1) [sphus]Mundlak_age = 0
( 2) [sphus]Mundlak_thinc = 0
( 3) [sphus]Mundlak_lifesat = 0
( 4) [sphus]Mundlak_yedu = 0
( 5) [sphus]Mundlak_age2 = 0
( 6) [sphus]Mundlak_thinc2 = 0
( 7) [sphus]Mundlak_lifesat2 = 0
( 8) [sphus]Mundlak_yedu2 = 0
( 9) [sphus]Mundlak_household_Single = 0
(10) [sphus]Mundlak_household_Couple = 0
(11) [sphus]o.Mundlak_household_Multiple = 0
(12) [sphus]Mundlak_covid_0 = 0
(13) [sphus]o.Mundlak_covid_1 = 0
(14) [sphus]Mundlak_cjs_Not = 0
(15) [sphus]Mundlak_cjs_Retired = 0
(16) [sphus]Mundlak_cjs_Employed = 0
(17) [sphus]Mundlak_cjs_Unemployed = 0
(18) [sphus]Mundlak_cjs_Permanently = 0
(19) [sphus]Mundlak_cjs_Homemaker = 0
(20) [sphus]o.Mundlak_cjs_Other = 0
(21) [sphus]Mundlak_gali_Not = 0
(22) [sphus]o.Mundlak_gali_Limited = 0
(23) [sphus]Mundlak_rhfo_None = 0
(24) [sphus]Mundlak_rhfo_At = 0
(25) [sphus]o.Mundlak_rhfo_Not = 0
(26) [sphus]Mundlak_ghfo_None = 0
(27) [sphus]Mundlak_ghfo_At = 0
(28) [sphus]o.Mundlak_ghfo_Not = 0
(29) [sphus]Mundlak_naly_None = 0
(30) [sphus]o.Mundlak_naly_At = 0
(31) [sphus]o.Mundlak_naly_Not = 0
(32) [sphus]Mundlak_i1 = 0
(33) [sphus]o.Mundlak_i2 = 0
(34) [sphus]o.Mundlak_i3 = 0
(35) [sphus]Mundlak_i4 = 0
(36) [sphus]o.Mundlak_i5 = 0
(37) [sphus]o.Mundlak_i6 = 0
(38) [sphus]o.Mundlak_i7 = 0
(39) [sphus]o.Mundlak_i8 = 0
(40) [sphus]o.Mundlak_i9 = 0
Constraint 11 dropped
Constraint 13 dropped
Constraint 20 dropped
Constraint 22 dropped
Constraint 25 dropped
Constraint 28 dropped

```

Constraint 30 dropped
 Constraint 31 dropped
 Constraint 33 dropped
 Constraint 34 dropped
 Constraint 36 dropped
 Constraint 37 dropped
 Constraint 38 dropped
 Constraint 39 dropped
 Constraint 40 dropped

chi2(25) = **2213.91**
 Prob > chi2 = **0.0000**

Click to Open File: [xto_1.doc](#)

note: **2.naly** omitted because of collinearity.

Iteration 0: Log pseudolikelihood = **-2.713e+08** (not concave)
 Iteration 1: Log pseudolikelihood = **-1.668e+08** (not concave)
 Iteration 2: Log pseudolikelihood = **-1.224e+08** (not concave)
 Iteration 3: Log pseudolikelihood = **-1.012e+08** (not concave)
 Iteration 4: Log pseudolikelihood = **-89161838** (not concave)
 Iteration 5: Log pseudolikelihood = **-85188065** (not concave)
 Iteration 6: Log pseudolikelihood = **-82460325** (not concave)
 Iteration 7: Log pseudolikelihood = **-81129483** (not concave)
 Iteration 8: Log pseudolikelihood = **-80188378** (not concave)
 Iteration 9: Log pseudolikelihood = **-79741769** (not concave)
 Iteration 10: Log pseudolikelihood = **-79269984** (not concave)
 Iteration 11: Log pseudolikelihood = **-79054061** (not concave)
 Iteration 12: Log pseudolikelihood = **-78847811** (not concave)
 Iteration 13: Log pseudolikelihood = **-78711331** (not concave)
 Iteration 14: Log pseudolikelihood = **-78601937** (not concave)
 Iteration 15: Log pseudolikelihood = **-78505596** (not concave)
 Iteration 16: Log pseudolikelihood = **-78414617** (not concave)
 Iteration 17: Log pseudolikelihood = **-78302097** (not concave)
 Iteration 18: Log pseudolikelihood = **-77969576** (not concave)
 Iteration 19: Log pseudolikelihood = **-77889021** (not concave)
 Iteration 20: Log pseudolikelihood = **-77826999** (not concave)
 Iteration 21: Log pseudolikelihood = **-77673538** (not concave)
 Iteration 22: Log pseudolikelihood = **-77647453** (not concave)
 Iteration 23: Log pseudolikelihood = **-77626508** (not concave)
 Iteration 24: Log pseudolikelihood = **-77613695** (not concave)
 Iteration 25: Log pseudolikelihood = **-77612342** (not concave)
 Iteration 26: Log pseudolikelihood = **-77612331** (not concave)
 Iteration 27: Log pseudolikelihood = **-77612331** (not concave)

Stereotype logistic regression

Number of obs = **29,989**

Log pseudolikelihood = **-77612331**

Wald chi2(35) = **2285.51**

Prob > chi2 = **0.0000**

(1) **[phi1_1]_cons = 1**

	sphus	Coefficient	Robust std. err.	z	P> z	[95% con
> f. interval]						
> .529931	age	.2783019	.1283845	2.17	0.030	.0266729
> .00029	c.age#c.age	-.0014336	.0008794	-1.63	0.103	-.0031573
> .0780874	yedu	.0023338	.0386505	0.06	0.952	-.0734198
> -.0027316	c.yedu#c.yedu	-.0062583	.0017994	-3.48	0.001	-.009785
> -.2542729	thinc	-.4770763	.1136773	-4.20	0.000	-.6998797
	c.thinc#c.thinc	.0724356	.0519762	1.39	0.163	-.0294358

>	.174307						
		gender					
		Female	.0290604	.1078437	0.27	0.788	-.1823095
>	.2404302						
		household					
		Couple, both responding	.3602395	.1423817	2.53	0.011	.0811766
>	.6393024						
		Multiple, at least 1 responding	.6773802	.1945684	3.48	0.000	.2960331
>	1.058727						
		cjs					
		Not applicable	-.0661864	.8714066	-0.08	0.939	-1.774112
>	1.641739						
		Retired	.5890243	.2098232	2.81	0.005	.1777785
>	1.00027						
		Unemployed	-.2089691	.4115225	-0.51	0.612	-1.015538
>	.5976002						
		Permanently sick or disabled	3.064677	.4472787	6.85	0.000	2.188027
>	3.941328						
		Homemaker	.4063403	.2430296	1.67	0.095	-.0699889
>	.8826695						
		Other	-1.748869	.4881406	-3.58	0.000	-2.705607
>	-.7921311						
		gali					
		Limited	5.374387	.1545342	34.78	0.000	5.071505
>	5.677268						
		lifesat	-.5121036	.1896166	-2.70	0.007	-.8837452
>	-.1404619						
		c.lifesat#c.lifesat	-.0200855	.0133101	-1.51	0.131	-.0461729
>	.0060019						
		naly					
		At least 1	-.9332433	.1402221	-6.66	0.000	-1.208074
>	-.6584131						
		Not applicable	0	(omitted)			
		rhfo					
		At least 1	1.225787	.1767183	6.94	0.000	.8794252
>	1.572148						
		ghfo					
		At least 1	-.4783553	.1544294	-3.10	0.002	-.7810314
>	-.1756792						
		Not applicable	-.0384871	.1243055	-0.31	0.757	-.2821214
>	.2051473						
		rhfo#ghfo					
		At least 1#At least 1	-1.055715	.3366167	-3.14	0.002	-1.715471
>	-.3959583						
		At least 1#Not applicable	-.0789618	.2856801	-0.28	0.782	-.6388845
>	.480961						
		country					
		Austria	-.2781977	.1940272	-1.43	0.152	-.658484
>	.1020886						
		Germany	1.073214	.2209932	4.86	0.000	.6400756
>	1.506353						
		Netherlands	-.179295	.2085884	-0.86	0.390	-.5881209
>	.2295308						
		Spain	1.967631	.2245366	8.76	0.000	1.527547
>	2.407715						
		Italy	1.017193	.2145038	4.74	0.000	.5967731
>	1.437613						
		France	.921428	.1933072	4.77	0.000	.5425528
>	1.300303						
		Denmark	-1.290782	.2414446	-5.35	0.000	-1.764005
>	-.8175592						
		Switzerland	-.6372098	.2193225	-2.91	0.004	-1.067074

>	-.2073456							
		Belgium		-.0268551	.1937898	-0.14	0.890	-.4066761
>	.3529658							
		Czech Republic		.7891246	.2334864	3.38	0.001	.3314996
>	1.24675							
		Slovenia		1.433844	.2796131	5.13	0.000	.8858124
>	1.981876							
		/phi1_1		1	(constrained)			
		/phi1_2		.8518712	.0178137	47.82	0.000	.8169571
>	.8867853							
		/phi1_3		.6267977	.0133346	47.01	0.000	.6006623
>	.6529331							
		/phi1_4		.3455562	.0119697	28.87	0.000	.3220961
>	.3690163							
		/phi1_5		0	(base outcome)			
		/theta1		9.763527	4.70093	2.08	0.038	.5498736
>	18.97718							
		/theta2		9.703074	4.009778	2.42	0.016	1.844054
>	17.56209							
		/theta3		8.96162	2.967966	3.02	0.003	3.144513
>	14.77873							
		/theta4		5.721882	1.654928	3.46	0.001	2.478282
>	8.965482							
		/theta5		0	(base outcome)			

(sphus=Poor is the base outcome)

results_impl_slo_1.xls

dir : seeout

note: **2.naly** omitted because of collinearity.

```

Iteration 0: Log pseudolikelihood = -2.418e+08 (not concave)
Iteration 1: Log pseudolikelihood = -2.029e+08 (not concave)
Iteration 2: Log pseudolikelihood = -1.301e+08 (not concave)
Iteration 3: Log pseudolikelihood = -98530387 (not concave)
Iteration 4: Log pseudolikelihood = -75452468 (not concave)
Iteration 5: Log pseudolikelihood = -67508549 (not concave)
Iteration 6: Log pseudolikelihood = -64314544 (not concave)
Iteration 7: Log pseudolikelihood = -62893416 (not concave)
Iteration 8: Log pseudolikelihood = -61738042 (not concave)
Iteration 9: Log pseudolikelihood = -61046948 (not concave)
Iteration 10: Log pseudolikelihood = -60521561 (not concave)
Iteration 11: Log pseudolikelihood = -60171917 (not concave)
Iteration 12: Log pseudolikelihood = -59871452 (not concave)
Iteration 13: Log pseudolikelihood = -59570554 (not concave)
Iteration 14: Log pseudolikelihood = -59422695 (not concave)
Iteration 15: Log pseudolikelihood = -58946252 (not concave)
Iteration 16: Log pseudolikelihood = -58749052 (not concave)
Iteration 17: Log pseudolikelihood = -58490995 (not concave)
Iteration 18: Log pseudolikelihood = -58237834 (not concave)
Iteration 19: Log pseudolikelihood = -58187595 (not concave)
Iteration 20: Log pseudolikelihood = -58149159 (not concave)
Iteration 21: Log pseudolikelihood = -58114770 (not concave)
Iteration 22: Log pseudolikelihood = -58082386 (not concave)
Iteration 23: Log pseudolikelihood = -58050428 (not concave)
Iteration 24: Log pseudolikelihood = -58019127 (not concave)
Iteration 25: Log pseudolikelihood = -57988725 (not concave)
Iteration 26: Log pseudolikelihood = -57959023 (not concave)
Iteration 27: Log pseudolikelihood = -57929925 (not concave)
Iteration 28: Log pseudolikelihood = -57901455 (not concave)
Iteration 29: Log pseudolikelihood = -57873663 (not concave)
Iteration 30: Log pseudolikelihood = -57685828 (not concave)
Iteration 31: Log pseudolikelihood = -57244223 (not concave)
Iteration 32: Log pseudolikelihood = -57185969 (not concave)
Iteration 33: Log pseudolikelihood = -57123091 (not concave)
Iteration 34: Log pseudolikelihood = -57109392 (not concave)
Iteration 35: Log pseudolikelihood = -57103333 (not concave)
Iteration 36: Log pseudolikelihood = -57102870 (not concave)

```

Iteration 37: Log pseudolikelihood = **-57102869**
 Iteration 38: Log pseudolikelihood = **-57102869**

Stereotype logistic regression

Number of obs = **22,899**
 Wald chi2(35) = **1205.15**
 Prob > chi2 = **0.0000**

Log pseudolikelihood = **-57102869**

(1) [phi1_1]_cons = 1

		sphus	Coefficient	Robust std. err.	z	P> z	[95% con
> f. interval]							
	age		.4577456	.1583817	2.89	0.004	.1473231
>	.768168						
	c.age#c.age		-.0027381	.0010623	-2.58	0.010	-.0048202
>	-.000656						
	yedu		-.0939242	.0433768	-2.17	0.030	-.1789413
>	-.0089072						
	c.yedu#c.yedu		-.0010787	.0019692	-0.55	0.584	-.0049383
>	.0027808						
	thinc		-.6610044	.1341785	-4.93	0.000	-.9239895
>	-.3980194						
	c.thinc#c.thinc		.1542501	.0631311	2.44	0.015	.0305155
>	.2779847						
	gender						
	Female		.0309071	.1237075	0.25	0.803	-.2115552
>	.2733693						
	household						
	Couple, both responding		.4463428	.156164	2.86	0.004	.1402669
>	.7524186						
	Multiple, at least 1 responding		.4581708	.2438417	1.88	0.060	-.0197501
>	.9360917						
	cjs						
	Not applicable		1.708125	.7852499	2.18	0.030	.1690631
>	3.247186						
	Retired		.9324669	.2914284	3.20	0.001	.3612779
>	1.503656						
	Unemployed		.2249463	.6520327	0.34	0.730	-1.053014
>	1.502907						
	Permanently sick or disabled		3.382052	.5617537	6.02	0.000	2.281035
>	4.483069						
	Homemaker		1.080066	.3393208	3.18	0.001	.4150096
>	1.745123						
	Other		.1269452	.5272896	0.24	0.810	-.9065234
>	1.160414						
	gali						
	Limited		5.084783	.1964355	25.89	0.000	4.699777
>	5.46979						
	lifesat		-.6441918	.1758837	-3.66	0.000	-.9889174
>	-.2994661						
	c.lifesat#c.lifesat		-.0067149	.0127188	-0.53	0.598	-.0316433
>	.0182135						
	naly						
	At least 1		-.6982245	.1603347	-4.35	0.000	-1.012475
>	-.3839742						
	Not applicable		0	(omitted)			
	rhfo						

>	1.45853	At least 1	1.071477	.1974801	5.43	0.000	.6844227
>		gheto					
>	-.4501258	At least 1	-.8117394	.1845001	-4.40	0.000	-1.173353
>	.1481307	Not applicable	-.130313	.1420657	-0.92	0.359	-.4087567
>		rhfo#gheto					
>	.1779896	At least 1#At least 1	-.6283435	.411402	-1.53	0.127	-1.434677
>	.767745	At least 1#Not applicable	.0774474	.3521991	0.22	0.826	-.6128501
>		country					
>	1.158297	Austria	.735777	.2155752	3.41	0.001	.3132574
>	2.529496	Germany	2.029267	.2552234	7.95	0.000	1.529038
>	.9826107	Netherlands	.5379411	.2268764	2.37	0.018	.0932716
>	2.865745	Spain	2.356874	.2596326	9.08	0.000	1.848003
>	2.227176	Italy	1.743464	.2467963	7.06	0.000	1.259752
>	2.165088	France	1.740442	.2166598	8.03	0.000	1.315797
>	.3351665	Denmark	-.1742441	.2599081	-0.67	0.503	-.6836547
>	.7562715	Switzerland	.288329	.2387506	1.21	0.227	-.1796136
>	.9525516	Belgium	.54333	.2087903	2.60	0.009	.1341085
>	2.132265	Czech Republic	1.59422	.274518	5.81	0.000	1.056174
>	2.04784	Slovenia	1.447462	.306321	4.73	0.000	.8470835
>		/phi1_1	1	(constrained)			
>	.9010989	/phi1_2	.8500632	.0260391	32.65	0.000	.7990274
>	.6670005	/phi1_3	.6283489	.0197206	31.86	0.000	.5896973
>	.3674963	/phi1_4	.3371336	.0154915	21.76	0.000	.3067708
>		/phi1_5	0	(base outcome)			
>	27.92372	/theta1	16.43787	5.860235	2.80	0.005	4.952026
>	25.15854	/theta2	15.33788	5.010629	3.06	0.002	5.51723
>	20.4726	/theta3	13.16822	3.726795	3.53	0.000	5.863834
>	11.80903	/theta4	7.834618	2.027797	3.86	0.000	3.860209
>		/theta5	0	(base outcome)			

(sphus=Poor is the base outcome)
results_impl_slo_1.xls
dir : seeout
note: 2.naly omitted because of collinearity.

```

Iteration 0: Log pseudolikelihood = -1.749e+08 (not concave)
Iteration 1: Log pseudolikelihood = -1.222e+08 (not concave)
Iteration 2: Log pseudolikelihood = -67082740 (not concave)
Iteration 3: Log pseudolikelihood = -52177117 (not concave)
Iteration 4: Log pseudolikelihood = -48915146 (not concave)
Iteration 5: Log pseudolikelihood = -48353530 (not concave)
Iteration 6: Log pseudolikelihood = -48089172 (not concave)
Iteration 7: Log pseudolikelihood = -47890454 (not concave)
Iteration 8: Log pseudolikelihood = -47723926 (not concave)
Iteration 9: Log pseudolikelihood = -47578979 (not concave)
Iteration 10: Log pseudolikelihood = -47458952 (not concave)
Iteration 11: Log pseudolikelihood = -47359013 (not concave)
Iteration 12: Log pseudolikelihood = -46937831 (not concave)
Iteration 13: Log pseudolikelihood = -46806128 (not concave)
Iteration 14: Log pseudolikelihood = -46771147 (not concave)
Iteration 15: Log pseudolikelihood = -46631063 (not concave)
Iteration 16: Log pseudolikelihood = -46578480 (not concave)
Iteration 17: Log pseudolikelihood = -46556134 (not concave)
Iteration 18: Log pseudolikelihood = -46553703 (not concave)
Iteration 19: Log pseudolikelihood = -46553516 (not concave)
Iteration 20: Log pseudolikelihood = -46553514 (not concave)
Iteration 21: Log pseudolikelihood = -46553514 (not concave)

```

Stereotype logistic regression

Number of obs = 18,545

Wald chi2(32) = 1120.99

Prob > chi2 = 0.0000

Log pseudolikelihood = -46553514

(1) [phi1_1]_cons = 1

	sphus	Coefficient	Robust std. err.	z	P> z	[95% con
> f. interval]						
> .9069326	age	.5669021	.1734882	3.27	0.001	.2268716
> -.0010702	c.age#c.age	-.0032951	.0011352	-2.90	0.004	-.00552
> .033794	yedu	-.0541019	.0448457	-1.21	0.228	-.1419978
> .0018018	c.yedu#c.yedu	-.002172	.0020275	-1.07	0.284	-.0061457
> -.2123615	thinc	-.5163438	.1550958	-3.33	0.001	-.8203261
> .1736053	c.thinc#c.thinc	.0327946	.0718435	0.46	0.648	-.108016
> .3452156	gender Female	.0734778	.1386443	0.53	0.596	-.19826
> .6874041	household Couple, both responding	.379906	.1568897	2.42	0.015	.0724079
> 1.148432	Multiple, at least 1 responding	.631859	.2635625	2.40	0.017	.115286
> 4.546725	cjs Not applicable	2.930202	.8247718	3.55	0.000	1.313679
> 2.4084	Retired	1.531857	.4472239	3.43	0.001	.6553141
> 2.031566	Unemployed	.4216813	.8213846	0.51	0.608	-1.188203
> 4.638559	Permanently sick or disabled	3.177264	.7455724	4.26	0.000	1.715969

>	2.677539	Homemaker	1.724198	.4864074	3.54	0.000	.7708573
>	2.762384	Other	1.697838	.5431458	3.13	0.002	.6332917
>	5.663623	gali Limited	5.26156	.2051377	25.65	0.000	4.859498
>	-.5986601	lifesat	-.9375456	.172904	-5.42	0.000	-1.276431
>	.0353189	c.lifesat#c.lifesat	.0107636	.0125284	0.86	0.390	-.0137917
>	.0513857	naly At least 1	-.2598831	.1588135	-1.64	0.102	-.5711519
		Not applicable	0	(omitted)			
>	1.612244	rhfo At least 1	1.267527	.1758791	7.21	0.000	.9228104
>	-.3378943	ghfo At least 1	-.7169315	.1933899	-3.71	0.000	-1.095969
>	-.3484005	rhfo#ghfo At least 1#At least 1	-1.042218	.3539949	-2.94	0.003	-1.736035
>	.9049486	country Austria	.484396	.2145716	2.26	0.024	.0638434
>	2.306827	Germany	1.807614	.2547057	7.10	0.000	1.3084
>	2.639882	Spain	2.129508	.2604	8.18	0.000	1.619133
>	2.171323	Italy	1.692463	.2443212	6.93	0.000	1.213602
>	2.021332	France	1.596097	.2169605	7.36	0.000	1.170862
>	.0460403	Denmark	-.4857553	.2713293	-1.79	0.073	-1.017551
>	.7373806	Switzerland	.2383389	.2546178	0.94	0.349	-.2607027
>	.6766472	Belgium	.2667835	.2091179	1.28	0.202	-.1430801
>	2.29659	Czech Republic	1.757884	.2748551	6.40	0.000	1.219178
>	2.451047	Slovenia	1.857424	.3028743	6.13	0.000	1.263801
>	.9307059	/phi1_1 /phi1_2	1 .880699	(constrained) .0255142	34.52	0.000	.8306921
>	.6980331	/phi1_3	.6599787	.0194159	33.99	0.000	.6219243
>	.3854855	/phi1_4 /phi1_5	.3529701 0	.0165898 (base outcome)	21.28	0.000	.3204546
>	34.57175	/theta1	21.65554	6.590024	3.29	0.001	8.739334
>	31.85177	/theta2	20.49158	5.796119	3.54	0.000	9.131399
>	25.76428	/theta3	17.1942	4.372571	3.93	0.000	8.62412
>	14.78012	/theta4	10.08059	2.397764	4.20	0.000	5.381063

/theta5	0 (base outcome)
---------	------------------

(sphus=Poor is the base outcome)

results_impl_slo_1.xls

dir : seeout

note: **2.naly** omitted because of collinearity.note: **2.ghfo** omitted because of collinearity.note: **0.rhfo#2.ghfo** identifies no observations in the sample.note: **1.rhfo#2.ghfo** identifies no observations in the sample.note: **2.rhfo#0.ghfo** identifies no observations in the sample.note: **2.rhfo#1.ghfo** identifies no observations in the sample.note: **2.rhfo#2.ghfo** omitted because of collinearity.

```

Iteration 0: Log pseudolikelihood = -2.038e+08 (not concave)
Iteration 1: Log pseudolikelihood = -1.013e+08 (not concave)
Iteration 2: Log pseudolikelihood = -61608569 (not concave)
Iteration 3: Log pseudolikelihood = -49037963 (not concave)
Iteration 4: Log pseudolikelihood = -46166887 (not concave)
Iteration 5: Log pseudolikelihood = -43791054 (not concave)
Iteration 6: Log pseudolikelihood = -42480506 (not concave)
Iteration 7: Log pseudolikelihood = -42196885 (not concave)
Iteration 8: Log pseudolikelihood = -42106253 (not concave)
Iteration 9: Log pseudolikelihood = -42047477 (not concave)
Iteration 10: Log pseudolikelihood = -42014631 (not concave)
Iteration 11: Log pseudolikelihood = -41984418
Iteration 12: Log pseudolikelihood = -41874054
Iteration 13: Log pseudolikelihood = -41818022 (not concave)
Iteration 14: Log pseudolikelihood = -41807620
Iteration 15: Log pseudolikelihood = -41783139 (not concave)
Iteration 16: Log pseudolikelihood = -41777807
Iteration 17: Log pseudolikelihood = -41777393
Iteration 18: Log pseudolikelihood = -41777389
Iteration 19: Log pseudolikelihood = -41777389

```

Stereotype logistic regression

Number of obs = **16,181**Wald chi2(33) = **839.15**Prob > chi2 = **0.0000**Log pseudolikelihood = **-41777389**(1) **[phi1_1]_cons = 1**

		sphus	Coefficient	Robust std. err.	z	P> z	[95% con
> f. interval]							
>	.6053184	age	.251385	.1805816	1.39	0.164	-.1025485
>	.0010117	c.age#c.age	-.0012733	.0011659	-1.09	0.275	-.0035584
>	.0592468	yedu	-.0325144	.0468178	-0.69	0.487	-.1242757
>	.001428	c.yedu#c.yedu	-.0027073	.0021099	-1.28	0.199	-.0068427
>	-.0226063	thinc	-.3388428	.1613481	-2.10	0.036	-.6550794
>	.0509207	c.thinc#c.thinc	-.1278266	.0911993	-1.40	0.161	-.3065739
>	.0681907	gender Female	-.2010845	.1373878	-1.46	0.143	-.4703597
>	.4393748	household Couple, both responding	.1468225	.1492641	0.98	0.325	-.1457298

Multiple, at least 1 responding		.1900239	.2722369	0.70	0.485	-.3435505
> .7235984						
	cjs					
> 4.303446	Not applicable	2.936704	.6973304	4.21	0.000	1.569962
> 2.318777	Retired	1.256984	.5417411	2.32	0.020	.1951915
> 3.960651	Unemployed	1.210123	1.403356	0.86	0.389	-1.540405
> 6.148885	Permanently sick or disabled	4.477026	.8530053	5.25	0.000	2.805166
> 2.663662	Homemaker	1.535484	.575612	2.67	0.008	.407305
> 2.362204	Other	.9381518	.7265706	1.29	0.197	-.4859005
	gali					
> 5.187257	Limited	4.767638	.2140953	22.27	0.000	4.348019
> .1139541	lifesat	-.2934772	.2078769	-1.41	0.158	-.7009084
	c.lifesat#c.lifesat					
> .0024903		-.0265101	.0147964	-1.79	0.073	-.0555105
	naly					
> -.0652059	At least 1	-.3928734	.1671803	-2.35	0.019	-.7205408
	Not applicable	0	(omitted)			
	rhfo					
> 1.217746	At least 1	.7634538	.231786	3.29	0.001	.3091616
> .5738068	Not applicable	.0828718	.2504816	0.33	0.741	-.4080632
	ghfo					
> -.2520114	At least 1	-.7110069	.2341857	-3.04	0.002	-1.170002
	Not applicable	0	(omitted)			
	rhfo#ghfo					
	None#Not applicable	0	(empty)			
> 1.625675	At least 1#At least 1	.7946569	.4239967	1.87	0.061	-.0363613
	At least 1#Not applicable	0	(empty)			
	Not applicable#None	0	(empty)			
	Not applicable#At least 1	0	(empty)			
	Not applicable#Not applicable	0	(omitted)			
	country					
> 1.141013	Austria	.6992689	.225384	3.10	0.002	.2575242
> 2.130358	Germany	1.611292	.2648347	6.08	0.000	1.092225
> 2.584431	Spain	2.062706	.266191	7.75	0.000	1.540982
> 2.062183	Italy	1.576033	.2480402	6.35	0.000	1.089883
> 1.515211	France	1.080316	.2218889	4.87	0.000	.6454222
> -.2374686	Denmark	-.7632991	.2682858	-2.85	0.004	-1.28913
> .5318548	Switzerland	.0684241	.2364486	0.29	0.772	-.3950067
> .4189466	Belgium	.0068422	.2102612	0.03	0.974	-.4052623
> .7018498	Czech Republic	.2349254	.2382311	0.99	0.324	-.231999
> 2.153125	Slovenia	1.597429	.2835237	5.63	0.000	1.041733

	/phi1_1	1	(constrained)			
	/phi1_2	.917454	.0315042	29.12	0.000	.8557069
>	.9792012					
	/phi1_3	.6684633	.0229569	29.12	0.000	.6234687
>	.713458					
	/phi1_4	.3439874	.0198477	17.33	0.000	.3050865
>	.3828882					
	/phi1_5	0	(base outcome)			
	/theta1	11.05637	7.020998	1.57	0.115	-2.704537
>	24.81727					
	/theta2	11.4151	6.466612	1.77	0.078	-1.259225
>	24.08943					
	/theta3	10.11436	4.722903	2.14	0.032	.857636
>	19.37108					
	/theta4	6.018467	2.474409	2.43	0.015	1.168715
>	10.86822					
	/theta5	0	(base outcome)			

(sphus=Poor is the base outcome)

results_impl_slo_1.xls

dir : seeout

note: 2.naly omitted because of collinearity.

```

Iteration 0: Log pseudolikelihood = -2.196e+08 (not concave)
Iteration 1: Log pseudolikelihood = -1.313e+08 (not concave)
Iteration 2: Log pseudolikelihood = -74248439 (not concave)
Iteration 3: Log pseudolikelihood = -44848942 (not concave)
Iteration 4: Log pseudolikelihood = -35685178 (not concave)
Iteration 5: Log pseudolikelihood = -32344497 (not concave)
Iteration 6: Log pseudolikelihood = -30985091 (not concave)
Iteration 7: Log pseudolikelihood = -30355902 (not concave)
Iteration 8: Log pseudolikelihood = -30038279 (not concave)
Iteration 9: Log pseudolikelihood = -29856597 (not concave)
Iteration 10: Log pseudolikelihood = -29678591 (not concave)
Iteration 11: Log pseudolikelihood = -29544015 (not concave)
Iteration 12: Log pseudolikelihood = -29475168 (not concave)
Iteration 13: Log pseudolikelihood = -29421365 (not concave)
Iteration 14: Log pseudolikelihood = -29391034 (not concave)
Iteration 15: Log pseudolikelihood = -29361066 (not concave)
Iteration 16: Log pseudolikelihood = -29248805 (not concave)
Iteration 17: Log pseudolikelihood = -29193966 (not concave)
Iteration 18: Log pseudolikelihood = -29168949 (not concave)
Iteration 19: Log pseudolikelihood = -29150153 (not concave)
Iteration 20: Log pseudolikelihood = -29135145 (not concave)
Iteration 21: Log pseudolikelihood = -29121242 (not concave)
Iteration 22: Log pseudolikelihood = -29101858 (not concave)
Iteration 23: Log pseudolikelihood = -29081670 (not concave)
Iteration 24: Log pseudolikelihood = -29059768 (not concave)
Iteration 25: Log pseudolikelihood = -29023089 (not concave)
Iteration 26: Log pseudolikelihood = -28918800 (not concave)
Iteration 27: Log pseudolikelihood = -28842485 (not concave)
Iteration 28: Log pseudolikelihood = -28837510 (not concave)
Iteration 29: Log pseudolikelihood = -28827613 (not concave)
Iteration 30: Log pseudolikelihood = -28821090 (not concave)
Iteration 31: Log pseudolikelihood = -28816090 (not concave)
Iteration 32: Log pseudolikelihood = -28811293 (not concave)
Iteration 33: Log pseudolikelihood = -28803318 (not concave)
Iteration 34: Log pseudolikelihood = -28796040 (not concave)
Iteration 35: Log pseudolikelihood = -28699606 (not concave)
Iteration 36: Log pseudolikelihood = -28610536 (not concave)
Iteration 37: Log pseudolikelihood = -28603822 (not concave)
Iteration 38: Log pseudolikelihood = -28597592 (not concave)
Iteration 39: Log pseudolikelihood = -28585888 (not concave)
Iteration 40: Log pseudolikelihood = -28577111 (not concave)
Iteration 41: Log pseudolikelihood = -28570742 (not concave)
Iteration 42: Log pseudolikelihood = -28564152 (not concave)
Iteration 43: Log pseudolikelihood = -28421825 (not concave)

```

```

Iteration 44: Log pseudolikelihood = -28390596 (not concave)
Iteration 45: Log pseudolikelihood = -28381199 (not concave)
Iteration 46: Log pseudolikelihood = -28370828 (not concave)
Iteration 47: Log pseudolikelihood = -28363544 (not concave)
Iteration 48: Log pseudolikelihood = -28357883 (not concave)
Iteration 49: Log pseudolikelihood = -28352593 (not concave)
Iteration 50: Log pseudolikelihood = -28347083 (not concave)
Iteration 51: Log pseudolikelihood = -28341682 (not concave)
Iteration 52: Log pseudolikelihood = -28336422 (not concave)
Iteration 53: Log pseudolikelihood = -28331138 (not concave)
Iteration 54: Log pseudolikelihood = -28325833 (not concave)
Iteration 55: Log pseudolikelihood = -28320573 (not concave)
Iteration 56: Log pseudolikelihood = -28315351 (not concave)
Iteration 57: Log pseudolikelihood = -28310142 (not concave)
Iteration 58: Log pseudolikelihood = -28304947 (not concave)
Iteration 59: Log pseudolikelihood = -28299782 (not concave)
Iteration 60: Log pseudolikelihood = -28294642 (not concave)
Iteration 61: Log pseudolikelihood = -28289526 (not concave)
Iteration 62: Log pseudolikelihood = -28284432 (not concave)
Iteration 63: Log pseudolikelihood = -28279366 (not concave)
Iteration 64: Log pseudolikelihood = -28274325 (not concave)
Iteration 65: Log pseudolikelihood = -28269313 (not concave)
Iteration 66: Log pseudolikelihood = -28264326 (not concave)
Iteration 67: Log pseudolikelihood = -28259370 (not concave)
Iteration 68: Log pseudolikelihood = -28254441 (not concave)
Iteration 69: Log pseudolikelihood = -28249544 (not concave)
Iteration 70: Log pseudolikelihood = -28244675 (not concave)
Iteration 71: Log pseudolikelihood = -28239840 (not concave)
Iteration 72: Log pseudolikelihood = -28235034 (not concave)
Iteration 73: Log pseudolikelihood = -28230262 (not concave)
Iteration 74: Log pseudolikelihood = -28225522 (not concave)
Iteration 75: Log pseudolikelihood = -28220817 (not concave)
Iteration 76: Log pseudolikelihood = -28216145 (not concave)
Iteration 77: Log pseudolikelihood = -28211510 (not concave)
Iteration 78: Log pseudolikelihood = -28023204 (not concave)
Iteration 79: Log pseudolikelihood = -28005975 (not concave)
Iteration 80: Log pseudolikelihood = -27930124 (not concave)
Iteration 81: Log pseudolikelihood = -27916505 (not concave)
Iteration 82: Log pseudolikelihood = -27909126 (not concave)
Iteration 83: Log pseudolikelihood = -27905228 (not concave)
Iteration 84: Log pseudolikelihood = -27904758 (not concave)
Iteration 85: Log pseudolikelihood = -27904734 (not concave)
Iteration 86: Log pseudolikelihood = -27904734 (not concave)

```

Stereotype logistic regression

Number of obs = 10,710

Log pseudolikelihood = -27904734

Wald chi2(33) = 742.49

Prob > chi2 = 0.0000

(1) [phi1_1]_cons = 1

	sphus	Coefficient	Robust std. err.	z	P> z	[95% con
> f. interval]						
> 1.079471	age	.5819194	.2538573	2.29	0.022	.0843683
> -.0003063	c.age#c.age	-.0034196	.0015885	-2.15	0.031	-.006533
> .047856	yedu	-.0714992	.0608966	-1.17	0.240	-.1908544
> .0045679	c.yedu#c.yedu	-.0007585	.0027176	-0.28	0.780	-.0060848
> .0559375	thinc	-.3093281	.1863634	-1.66	0.097	-.6745937
	c.thinc#c.thinc	.0679432	.0966919	0.70	0.482	-.1215696

>	.2574559						
		gender					
		Female	-.0225822	.1776699	-0.13	0.899	-.3708088
>	.3256444						
		household					
		Couple, both responding	.5423103	.2031055	2.67	0.008	.1442309
>	.9403897						
		Multiple, at least 1 responding	.5322242	.3322402	1.60	0.109	-.1189546
>	1.183403						
		cjs					
		Not applicable	3.835177	1.25265	3.06	0.002	1.380029
>	6.290326						
		Retired	3.804245	1.057115	3.60	0.000	1.732338
>	5.876152						
		Unemployed	3.0505	2.51076	1.21	0.224	-1.870499
>	7.9715						
		Permanently sick or disabled	5.421936	1.35265	4.01	0.000	2.770791
>	8.073081						
		Homemaker	3.417055	1.085796	3.15	0.002	1.288935
>	5.545175						
		Other	3.484748	1.180293	2.95	0.003	1.171416
>	5.79808						
		gali					
		Limited	5.248198	.2890442	18.16	0.000	4.681682
>	5.814714						
		lifesat	-1.429491	.3099207	-4.61	0.000	-2.036924
>	-.8220574						
		c.lifesat#c.lifesat	.0435233	.0215932	2.02	0.044	.0012014
>	.0858452						
		naly					
		At least 1	-.859238	.2613007	-3.29	0.001	-1.371378
>	-.347098						
		Not applicable	0	(omitted)			
		rhfo					
		At least 1	.7999046	.2280776	3.51	0.000	.3528807
>	1.246928						
		ghfo					
		At least 1	-.5265209	.237606	-2.22	0.027	-.9922202
>	-.0608217						
		rhfo#ghfo					
		At least 1#At least 1	-.2714711	.4616975	-0.59	0.557	-1.176382
>	.6334393						
		country					
		Austria	.4741381	.2983601	1.59	0.112	-.110637
>	1.058913						
		Germany	1.165182	.3375733	3.45	0.001	.5035505
>	1.826814						
		Netherlands	-.1299795	.3186009	-0.41	0.683	-.7544257
>	.4944667						
		Spain	2.100034	.3615049	5.81	0.000	1.391497
>	2.808571						
		Italy	1.31889	.3562336	3.70	0.000	.6206851
>	2.017095						
		France	.9808278	.2841669	3.45	0.001	.4238709
>	1.537785						
		Denmark	-.4050981	.344718	-1.18	0.240	-1.080733
>	.2705368						
		Switzerland	-.2376019	.3435874	-0.69	0.489	-.9110209
>	.435817						
		Belgium	.0895587	.2939967	0.30	0.761	-.4866643
>	.6657817						
		Czech Republic	-.3038001	.3373253	-0.90	0.368	-.9649454

>	.3573453					
>	1.826377	Slovenia	1.093083	.3741362	2.92	0.003 .3597898
		/phi1_1	1 (constrained)			
>	.9210231	/phi1_2	.8457969	.0383814	22.04	0.000 .7705706
>	.7001945	/phi1_3	.6447201	.0283038	22.78	0.000 .5892456
>	.3747775	/phi1_4	.3300615	.0228147	14.47	0.000 .2853454
		/phi1_5	0 (base outcome)			
		/theta1	21.58333	10.10043	2.14	0.033 1.786845
>	41.37982	/theta2	19.91098	8.679908	2.29	0.022 2.898676
>	36.92329	/theta3	16.89764	6.581984	2.57	0.010 3.997192
>	29.7981	/theta4	9.583672	3.448471	2.78	0.005 2.824794
>	16.34255	/theta5	0 (base outcome)			

(sphus=Poor is the base outcome)

results_impl_slo_1.xls

dir : seeout

note: 2.naly omitted because of collinearity.

Iteration 0: Log pseudolikelihood = -1.702e+08 (not concave)
 Iteration 1: Log pseudolikelihood = -1.431e+08 (not concave)
 Iteration 2: Log pseudolikelihood = -96853287 (not concave)
 Iteration 3: Log pseudolikelihood = -51246739 (not concave)
 Iteration 4: Log pseudolikelihood = -35118944 (not concave)
 Iteration 5: Log pseudolikelihood = -31393076 (not concave)
 Iteration 6: Log pseudolikelihood = -29910099 (not concave)
 Iteration 7: Log pseudolikelihood = -29443774 (not concave)
 Iteration 8: Log pseudolikelihood = -29076616 (not concave)
 Iteration 9: Log pseudolikelihood = -28832985 (not concave)
 Iteration 10: Log pseudolikelihood = -28636722 (not concave)
 Iteration 11: Log pseudolikelihood = -28468730 (not concave)
 Iteration 12: Log pseudolikelihood = -28384028 (not concave)
 Iteration 13: Log pseudolikelihood = -28274560 (not concave)
 Iteration 14: Log pseudolikelihood = -28183938 (not concave)
 Iteration 15: Log pseudolikelihood = -28130405 (not concave)
 Iteration 16: Log pseudolikelihood = -28105390 (not concave)
 Iteration 17: Log pseudolikelihood = -28093354 (not concave)
 Iteration 18: Log pseudolikelihood = -28088330 (not concave)
 Iteration 19: Log pseudolikelihood = -28087603 (not concave)
 Iteration 20: Log pseudolikelihood = -28087595 (not concave)
 Iteration 21: Log pseudolikelihood = -28087595 (not concave)

Stereotype logistic regression

Log pseudolikelihood = -28087595

Number of obs = 11,204

Wald chi2(32) = 593.67

Prob > chi2 = 0.0000

(1) [phi1_1]_cons = 1

	sphus	Coefficient	Robust std. err.	z	P> z	[95% con
> f. interval]						
>	1.523585	age	.9241015	.3058646	3.02	0.003 .3246179
>	-.0016847	c.age#c.age	-.0053854	.0018881	-2.85	0.004 -.009086

>	.171238	yedu	.0412601	.0663165	0.62	0.534	-.0887179
>	.0014141	c.yedu#c.yedu	-.0043146	.0029229	-1.48	0.140	-.0100433
>	-.0301588	thinc	-.4088937	.1932356	-2.12	0.034	-.7876286
>	.1914237	c.thinc#c.thinc	.0055427	.094839	0.06	0.953	-.1803383
>	.4630455	gender Female	.1071054	.1816054	0.59	0.555	-.2488347
>	1.15564	household Couple, both responding	.7467594	.2086165	3.58	0.000	.3378786
>	1.293759	Multiple, at least 1 responding	.4833272	.413493	1.17	0.242	-.3271043
>	3.594171	cjs Not applicable	-.298025	1.985851	-0.15	0.881	-4.190221
>	3.346186	Retired	-.2875241	1.853968	-0.16	0.877	-3.921234
>	5.046587	Permanently sick or disabled	.9101685	2.110456	0.43	0.666	-3.22625
>	2.919689	Homemaker	-.753707	1.874216	-0.40	0.688	-4.427103
>	4.276417	Other	.4117824	1.971789	0.21	0.835	-3.452853
>	6.324209	gali Limited	5.721853	.3073305	18.62	0.000	5.119496
>	-.2531138	lifesat	-.83549	.2971362	-2.81	0.005	-1.417866
>	.0322242	c.lifesat#c.lifesat	-.0087439	.0209025	-0.42	0.676	-.0497121
>	-.4168673	naly At least 1	-.864223	.2282469	-3.79	0.000	-1.311579
		Not applicable	0	(omitted)			
>	1.262017	rhfo At least 1	.8285048	.2211837	3.75	0.000	.3949927
>	-.1104045	ghfo At least 1	-.6658686	.2834052	-2.35	0.019	-1.221333
>	1.145232	rhfo#ghfo At least 1#At least 1	.1509232	.5073096	0.30	0.766	-.8433852
>	1.626835	country Austria	1.023481	.3078395	3.32	0.001	.4201266
>	2.363955	Germany	1.647202	.3656972	4.50	0.000	.9304488
>	.5409364	Netherlands	-.1314238	.3430472	-0.38	0.702	-.803784
>	3.045989	Spain	2.297893	.3816889	6.02	0.000	1.549797
>	2.37742	Italy	1.676775	.3574789	4.69	0.000	.9761289
		France	1.098456	.3076829	3.57	0.000	.4954091

>	1.701504	Denmark	-.055143	.3674447	-0.15	0.881	-.7753214
>	.6650354	Switzerland	.3455392	.3715126	0.93	0.352	-.3826121
>	1.073691	Belgium	.2763524	.3010095	0.92	0.359	-.3136154
>	.8663202	Czech Republic	-.0245503	.3773951	-0.07	0.948	-.7642312
>	.7151306	Slovenia	1.476409	.3607385	4.09	0.000	.7693743
>	2.183443						
		/phil_1	1 (constrained)				
>	.9067073	/phil_2	.8351717	.0364984	22.88	0.000	.7636361
>	.6905052	/phil_3	.6335418	.0290635	21.80	0.000	.5765785
>	.4019449	/phil_4	.3551902	.0238549	14.89	0.000	.3084356
		/phil_5	0 (base outcome)				
		/theta1	35.02662	12.50732	2.80	0.005	10.51273
>	59.54051	/theta2	30.8629	10.36555	2.98	0.003	10.54679
>	51.17901	/theta3	25.28157	7.878271	3.21	0.001	9.840446
>	40.7227	/theta4	15.01989	4.408869	3.41	0.001	6.378669
>	23.66112	/theta5	0 (base outcome)				

(sphus=Poor is the base outcome)

results_impl_slo_1.xls

dir : seeout

name: The value of a helping hand

log: C:\Users\hansg\Documents\working\SHARE data\weighting\output_v6_1.smcl

log type: smcl

closed on: 9 Jun 2026, 19:29:50